EL-AZHARY SALAH-EDDINE

Smart Detector



Plan du presentation



Introduction



Cahier de charges



Problematique



Objectifs



Prototype



Conclusion



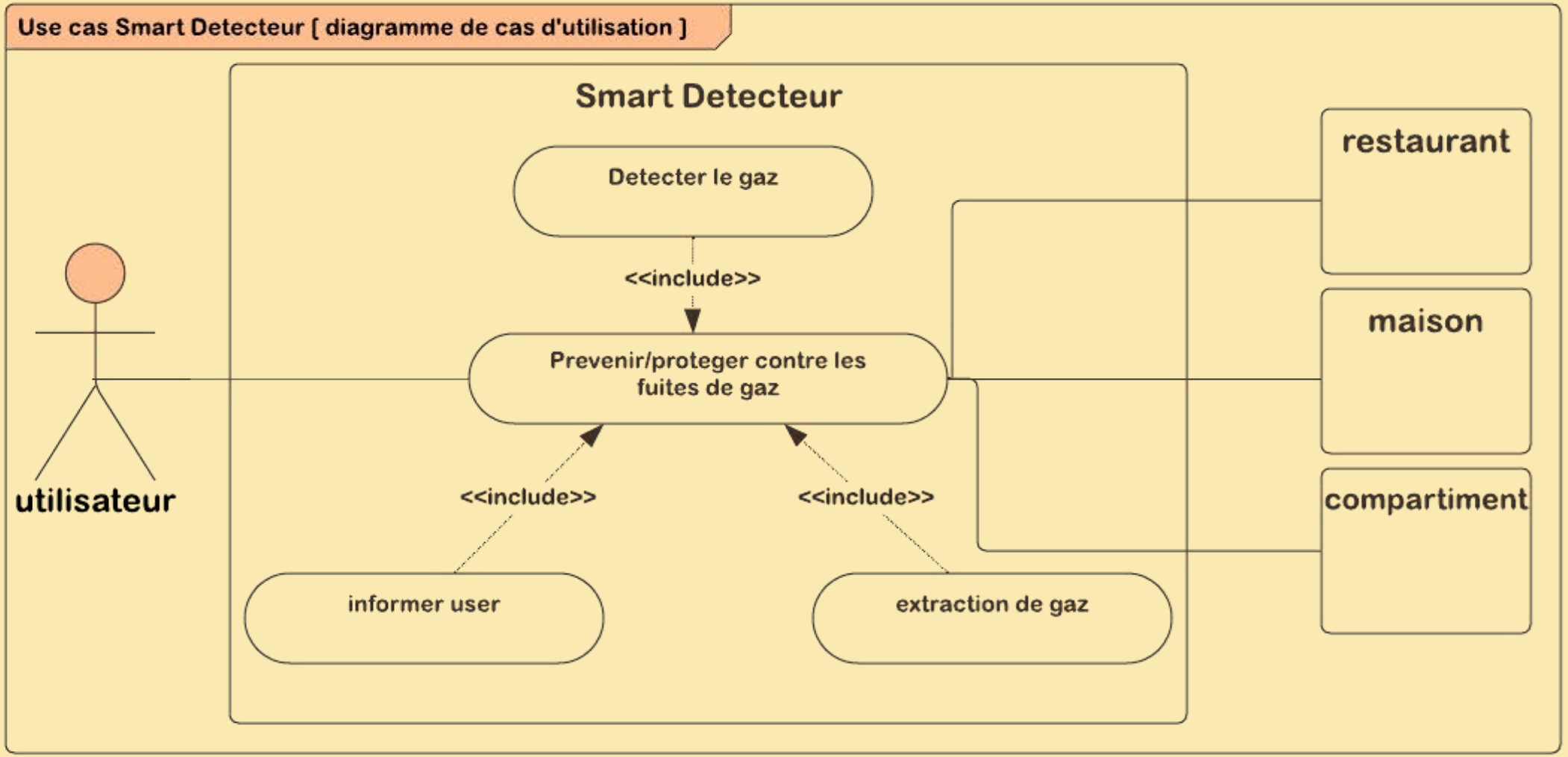
Figure 1: Décès/blessés suite à l'explosion d'une bonbonne de gaz dans le quartier Al-Charak

Casablanca (2020)



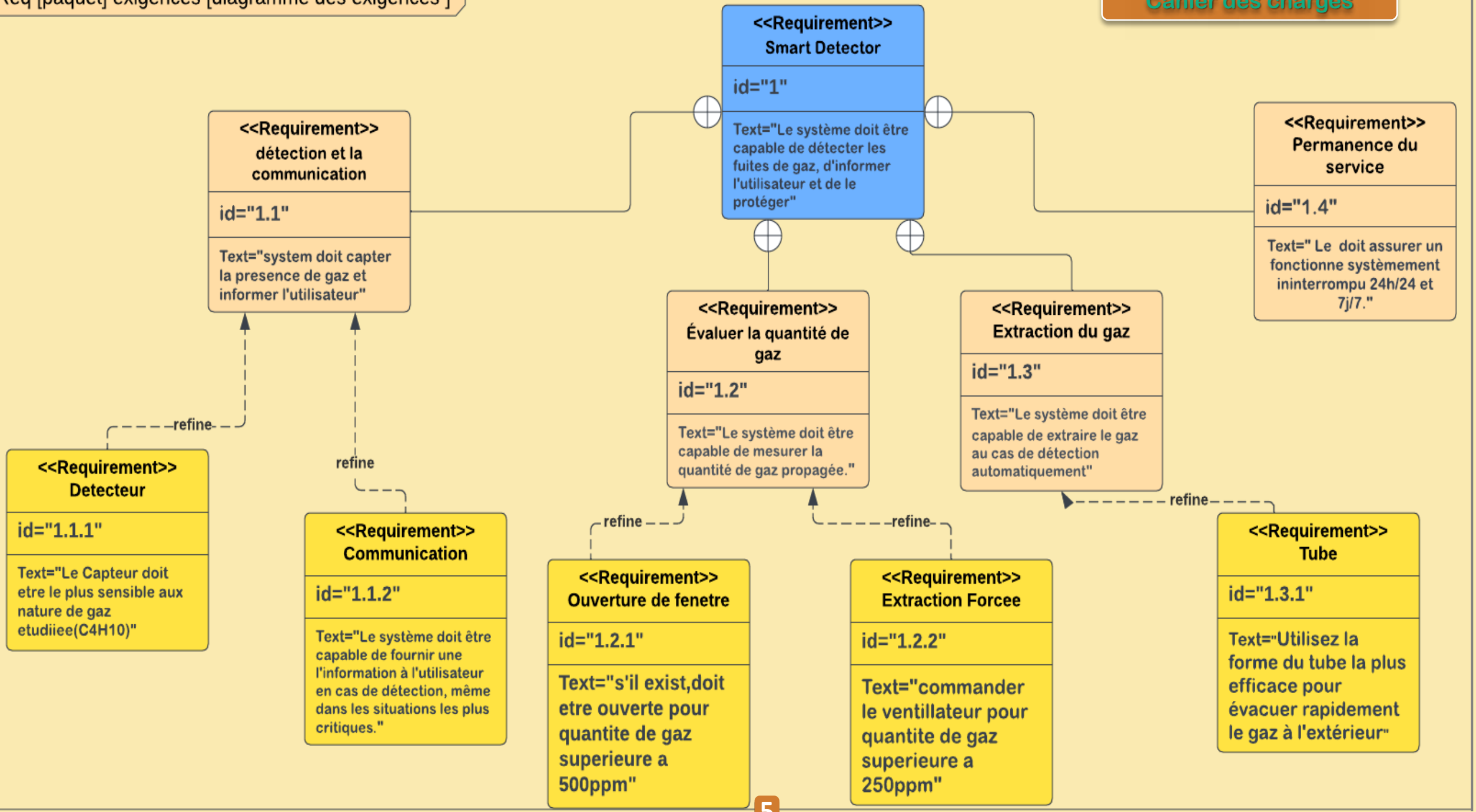
Figure 2: L'explosion rue de Trévise avait eu lieu en janvier 2019.

Paris (2019)


Use cas Smart Detecteur [diagramme de cas d'utilisation]
Smart Detecteur
Detecter le gaz
<<include>>
Prevenir/proteger contre les fuites de gaz
<<include>>
<<include>>
informer user
extraction de gaz
utilisateur
restaurant
maison
compartiment

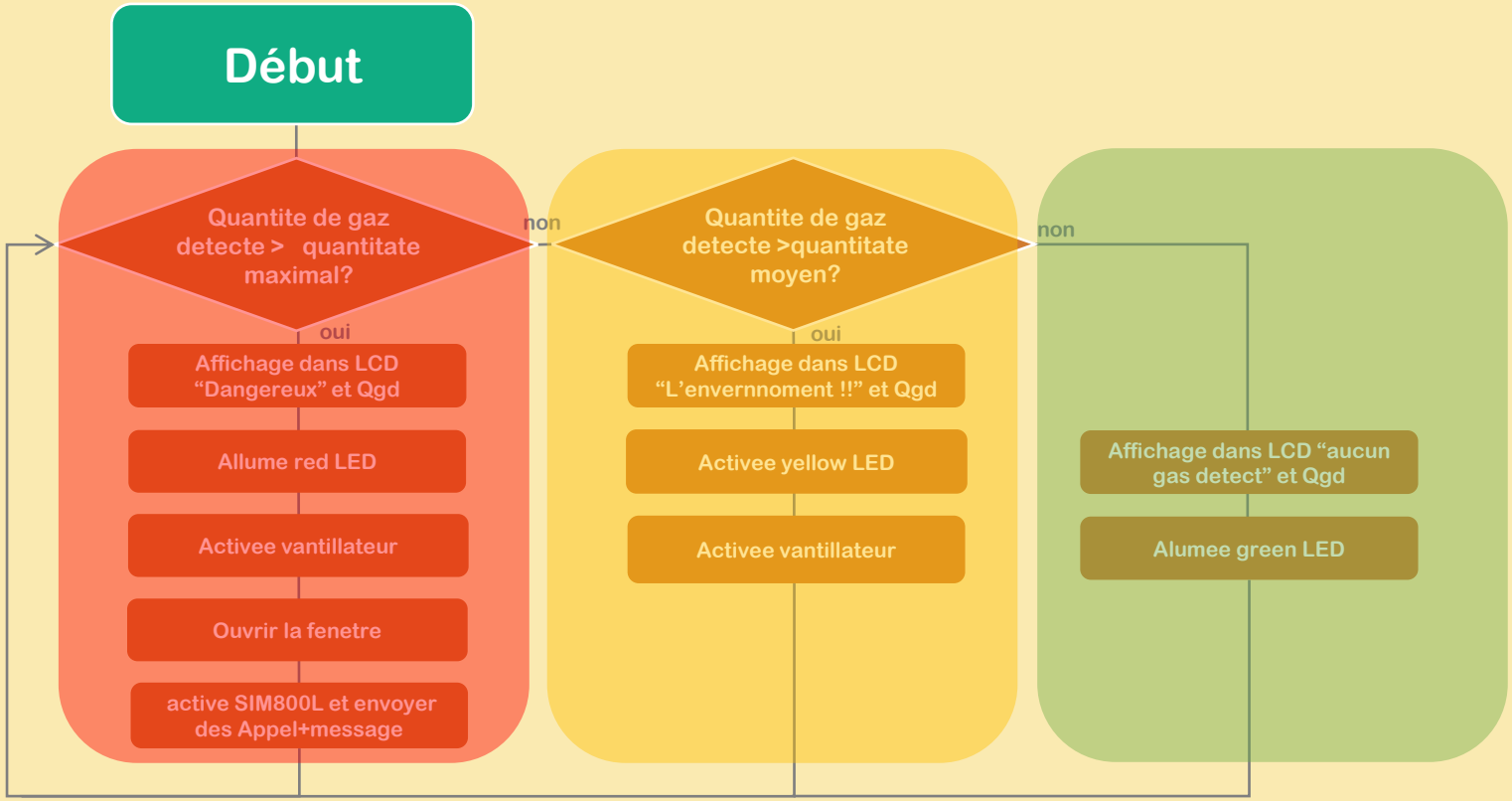
Req [paquet] exigences [diagramme des exigences]

Cahier des charges

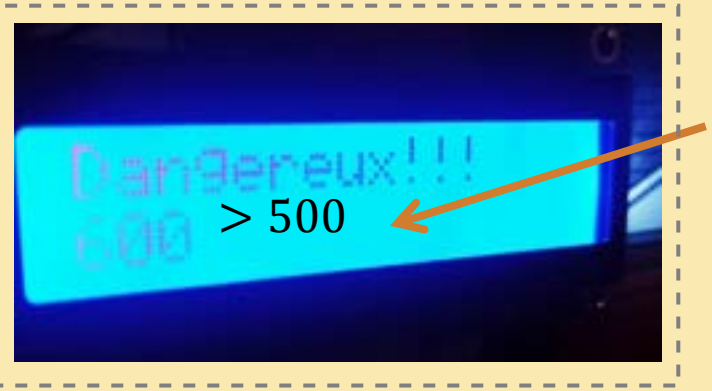




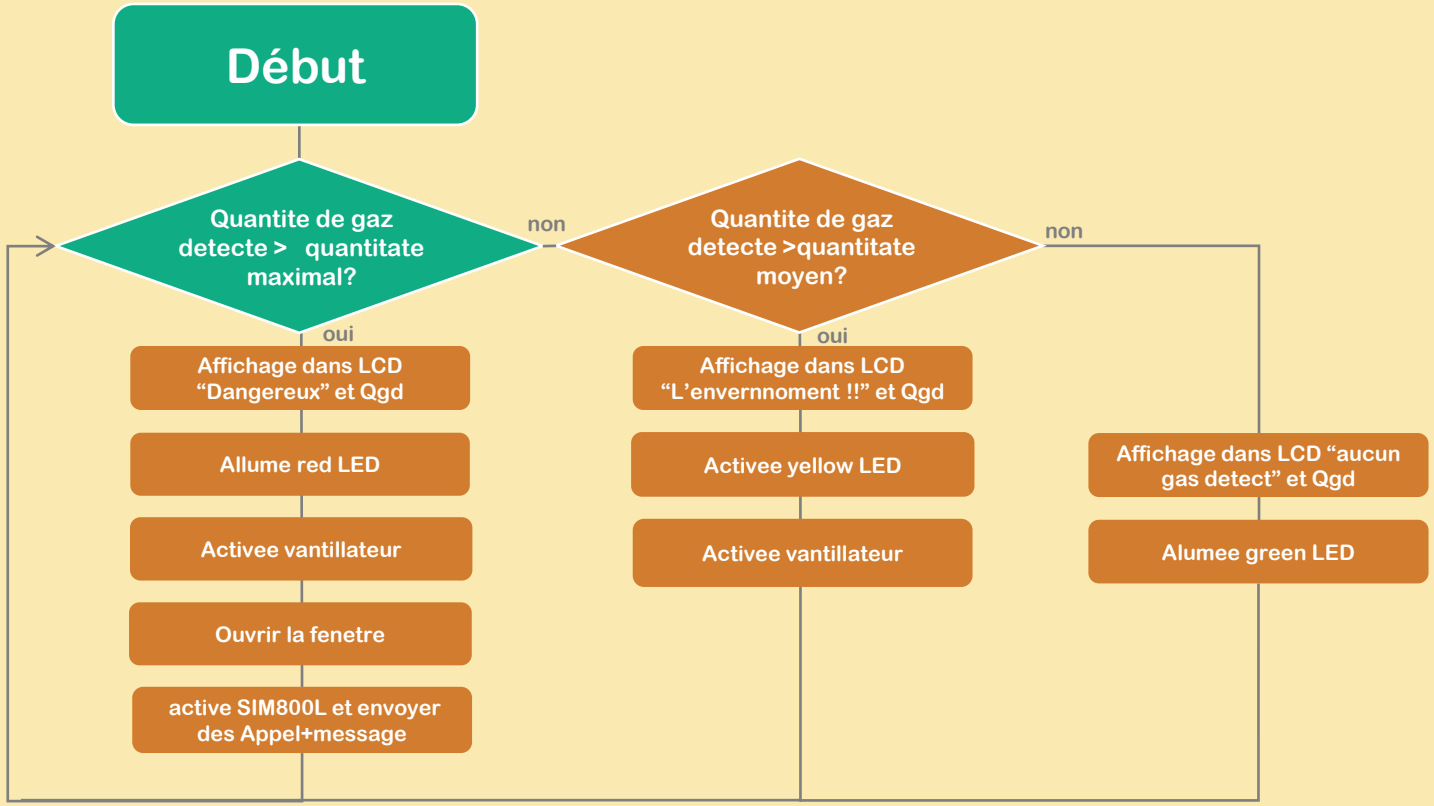
Comment peut-on detectee et extraire complétement le gaz avec la meilleure réception d'information par l'utilisateur?



Zones de fonctionnement



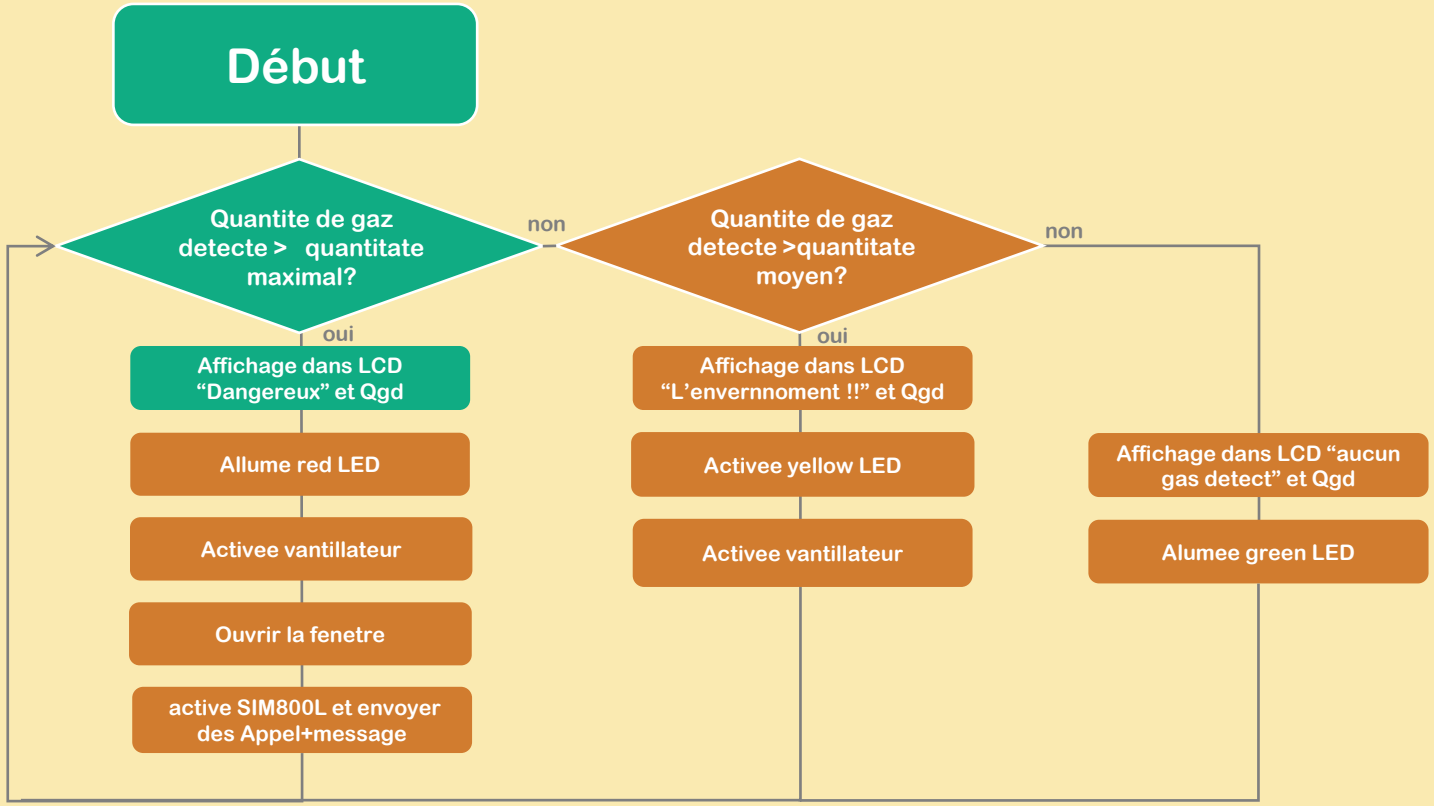
Prototype de l'operation



Algorigramme (fonctionnement)



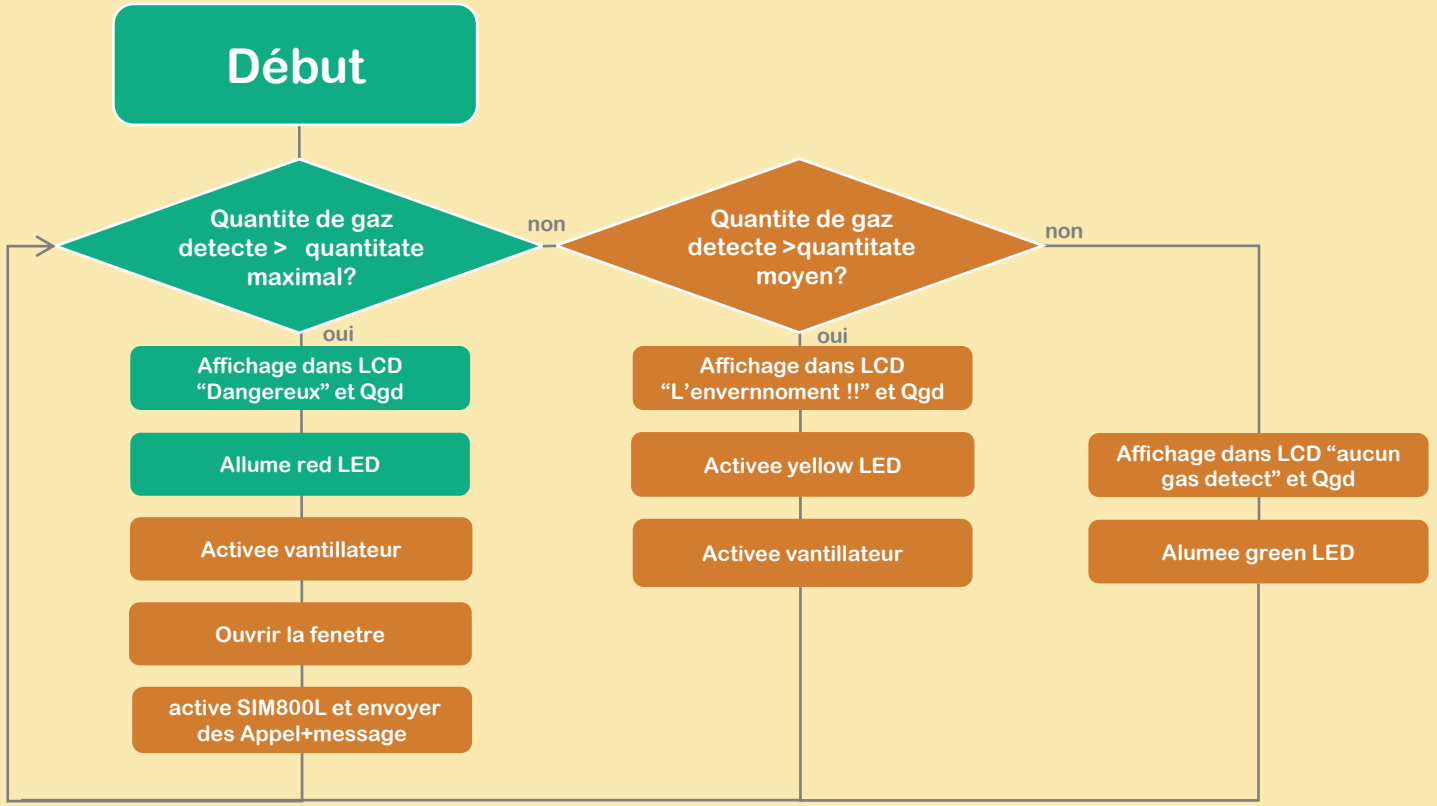
Prototype de l'operation



Algorigramme (fonctionnement)



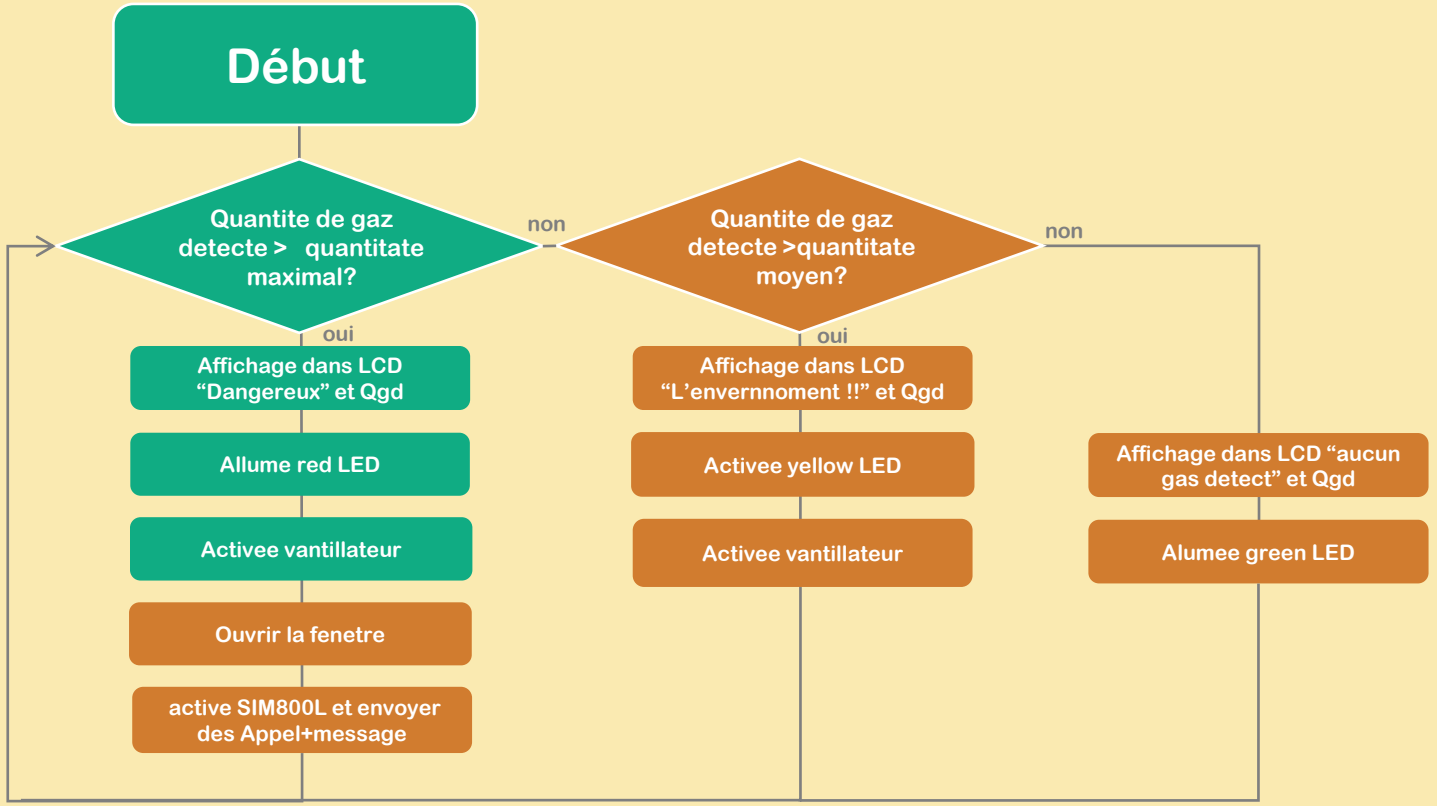
Prototype de l'operation



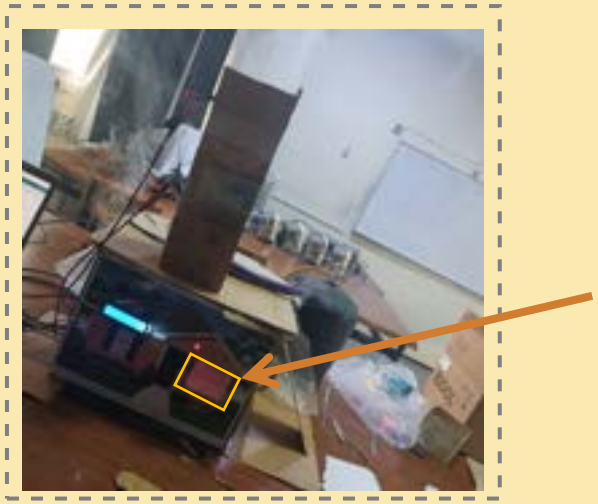
Algorigramme (fonctionnement)



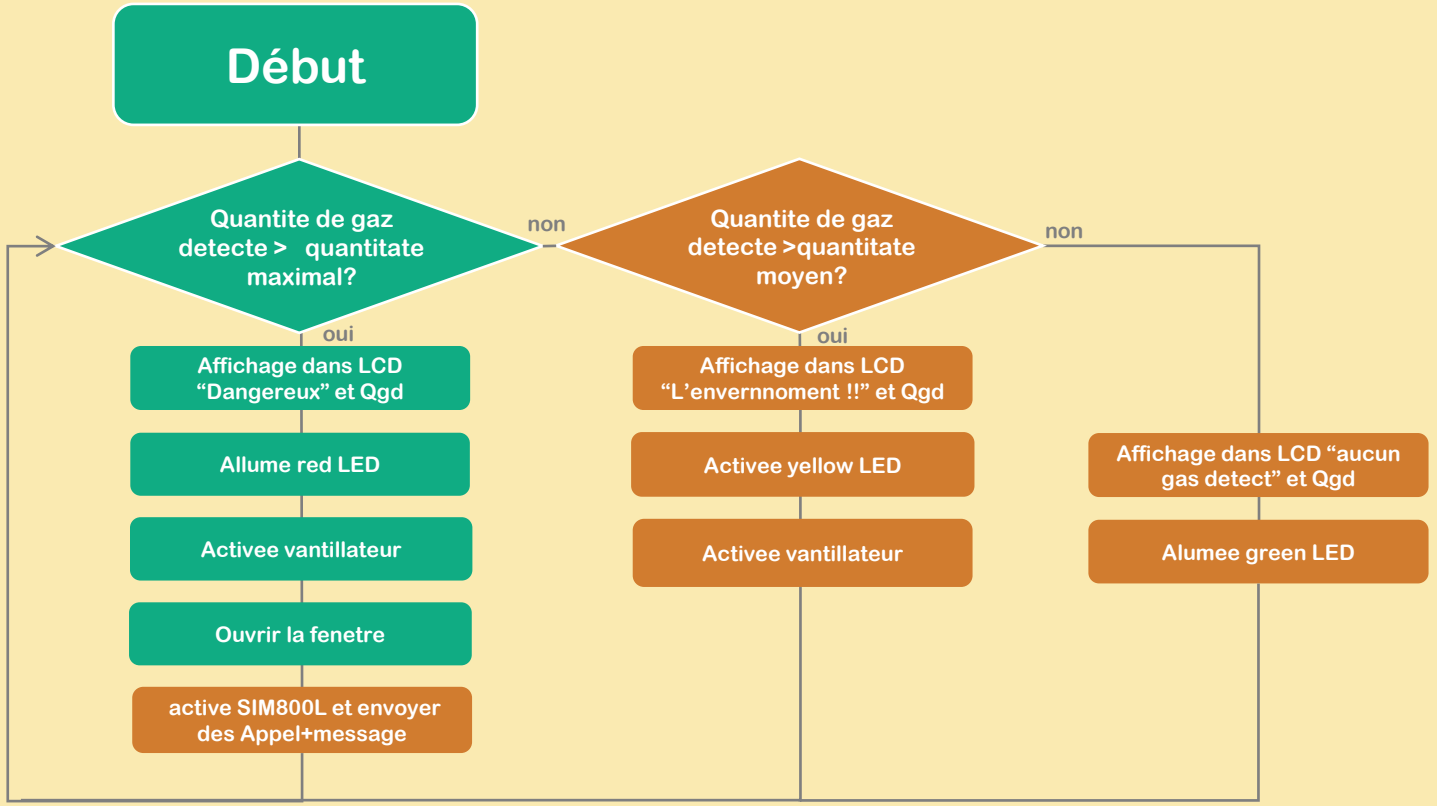
Prototype de l'operation



Algorigramme (fonctionnement)



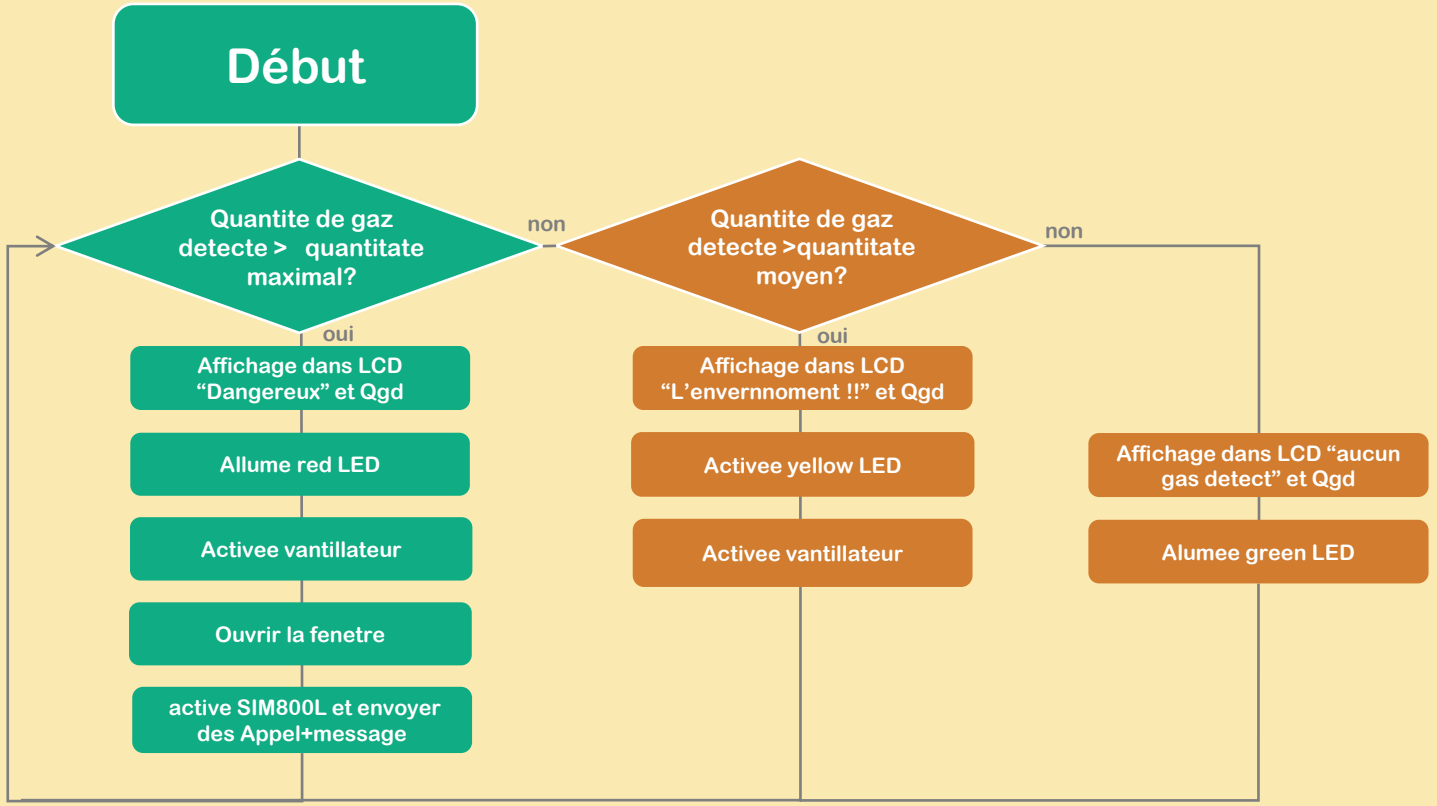
Prototype de l'operation



Algorigramme (fonctionnement)



Prototype de l'operation

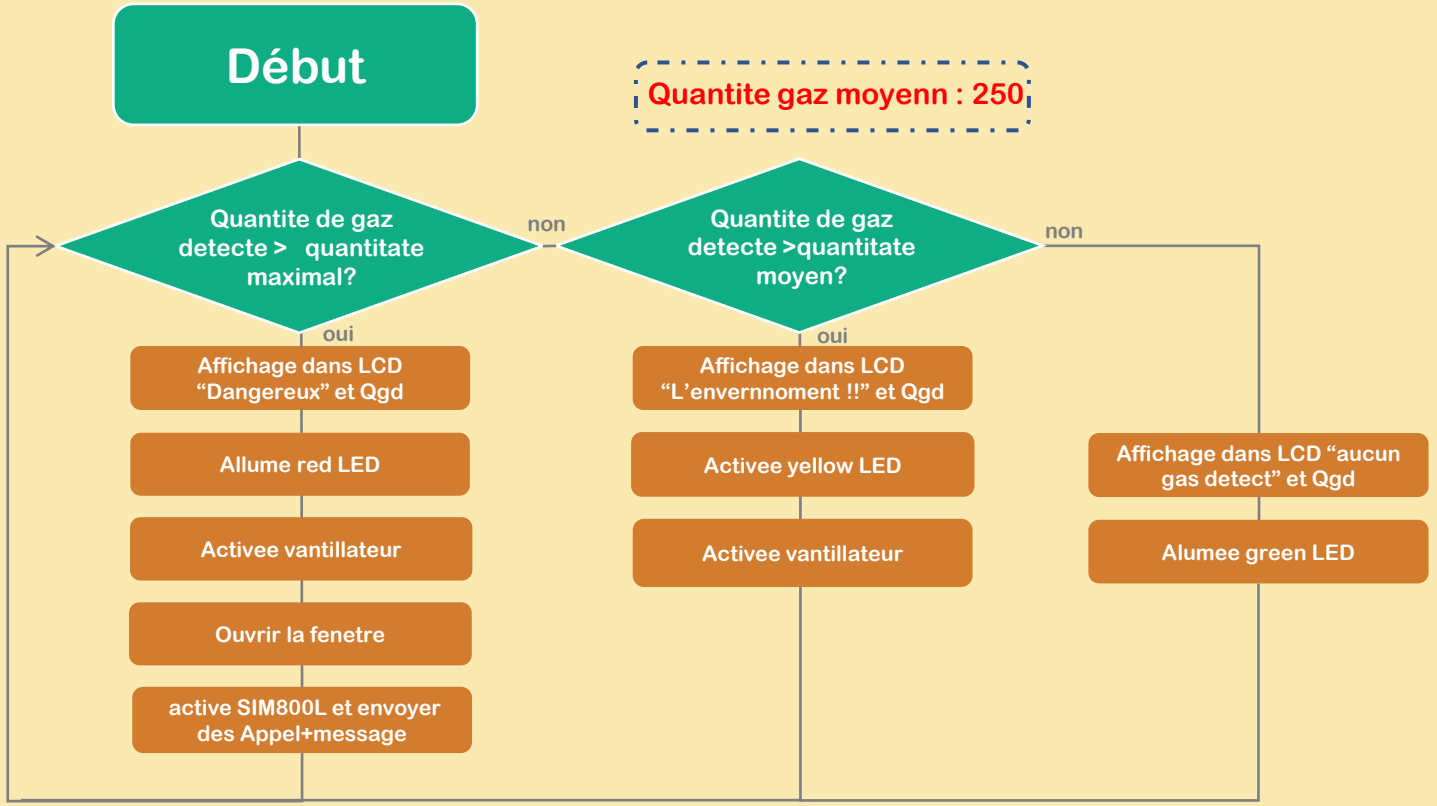


Algorigramme (fonctionnement)

$500 > 300 > 250$



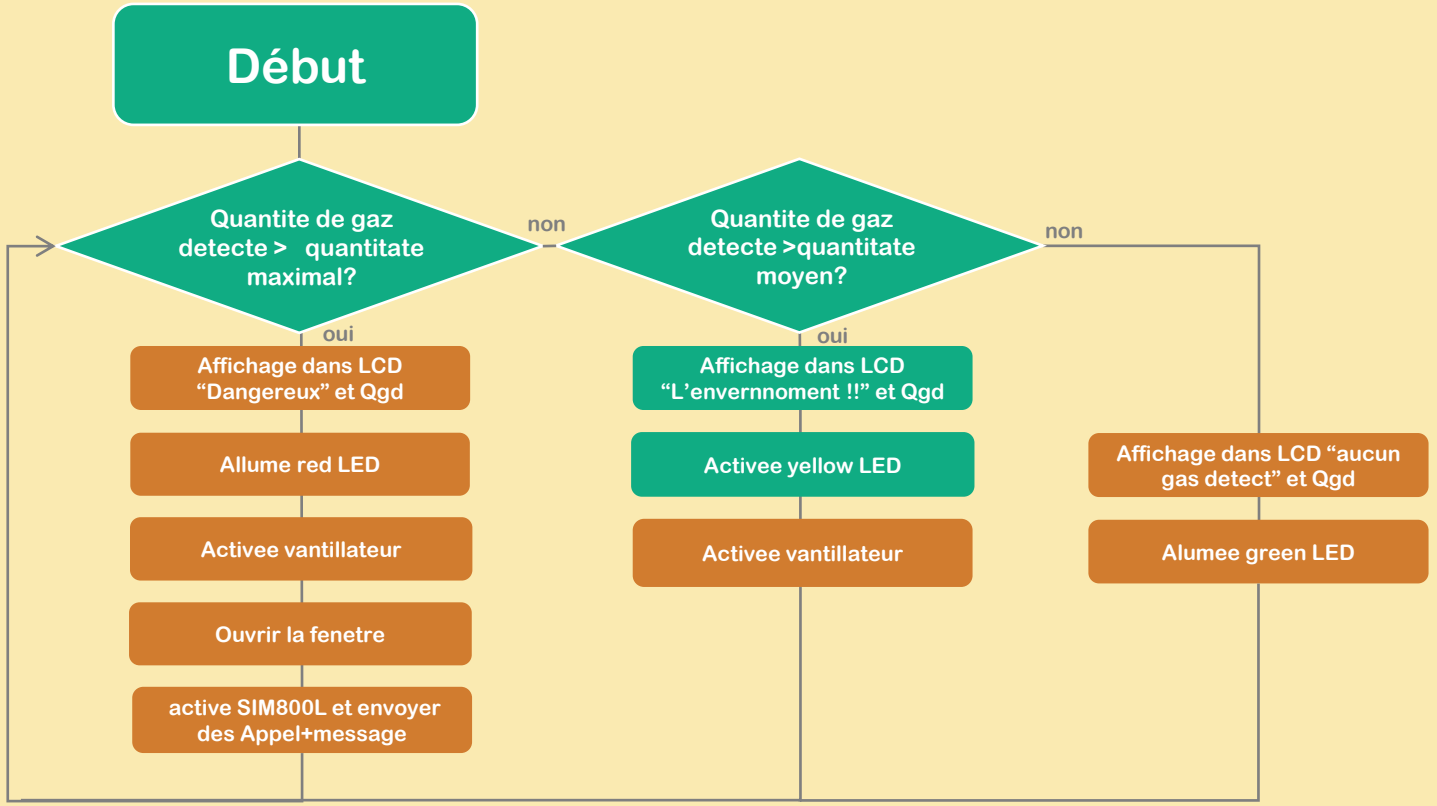
Prototype de l'operation



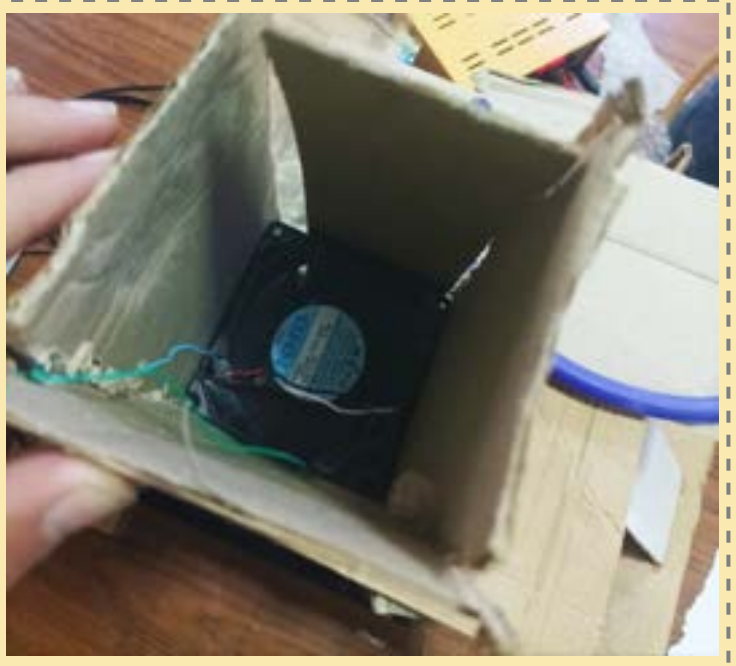
Algorigramme (fonctionnement)



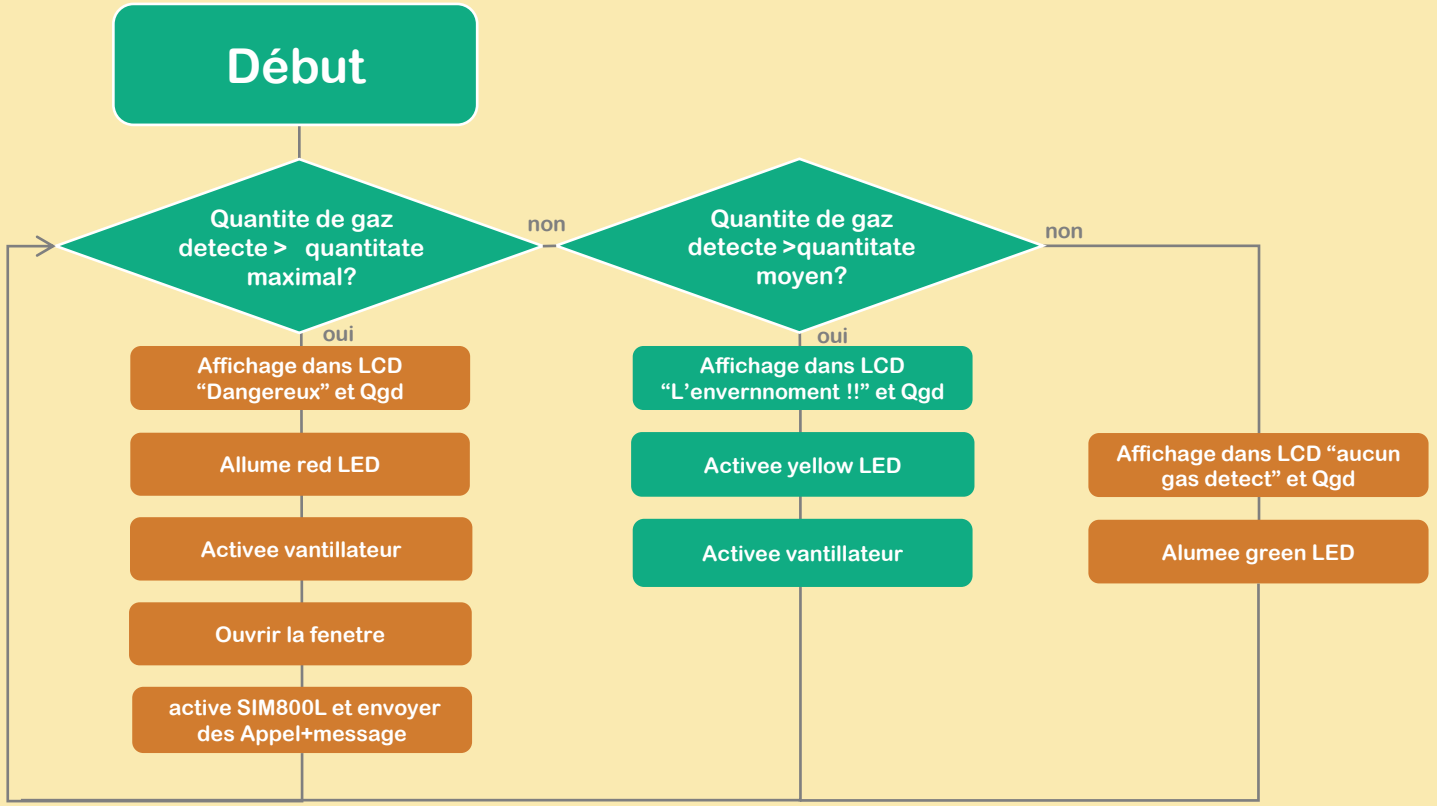
Prototype de l'operation



Algorigramme (fonctionnement)



Prototype de l'operation

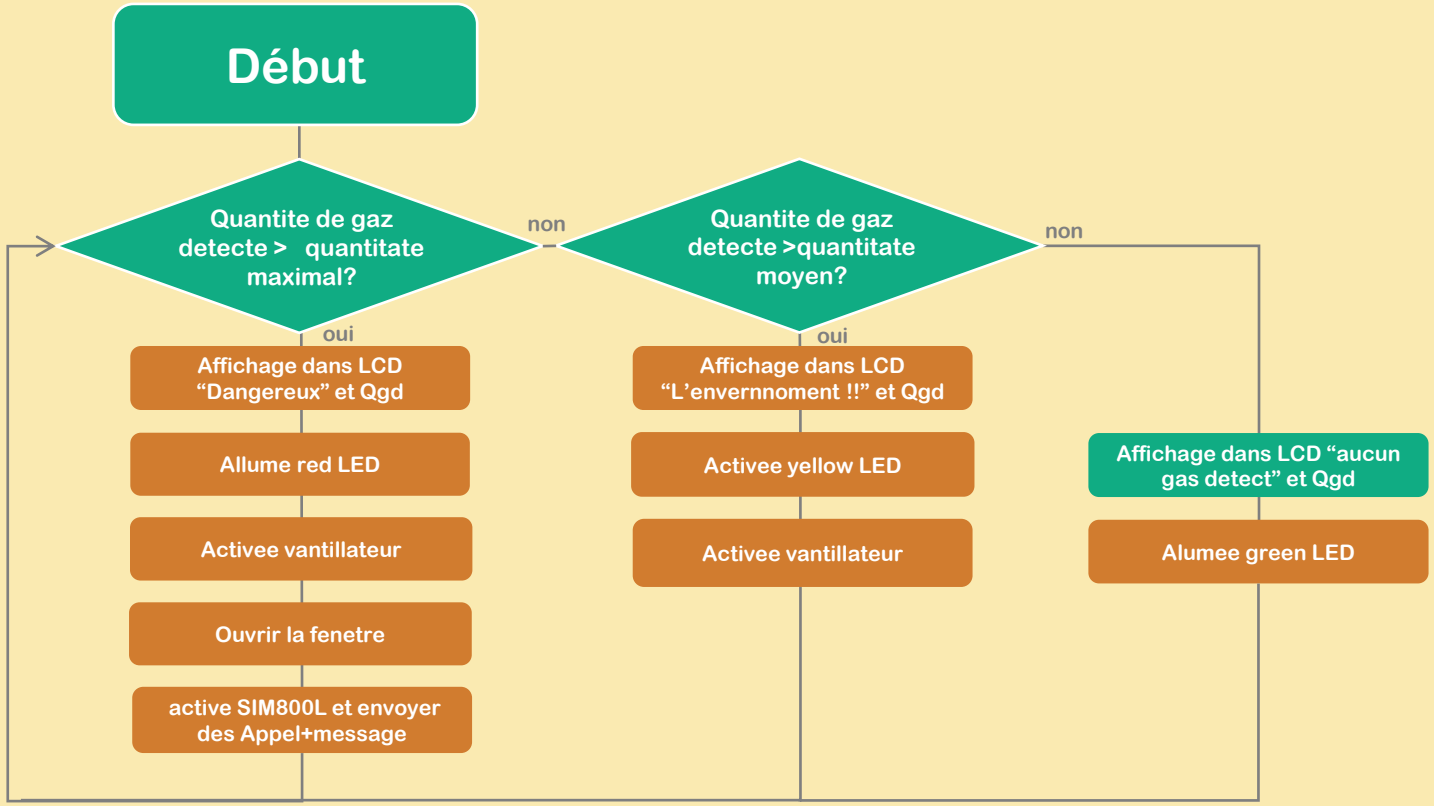


Algorigramme (fonctionnement)

250 > 20



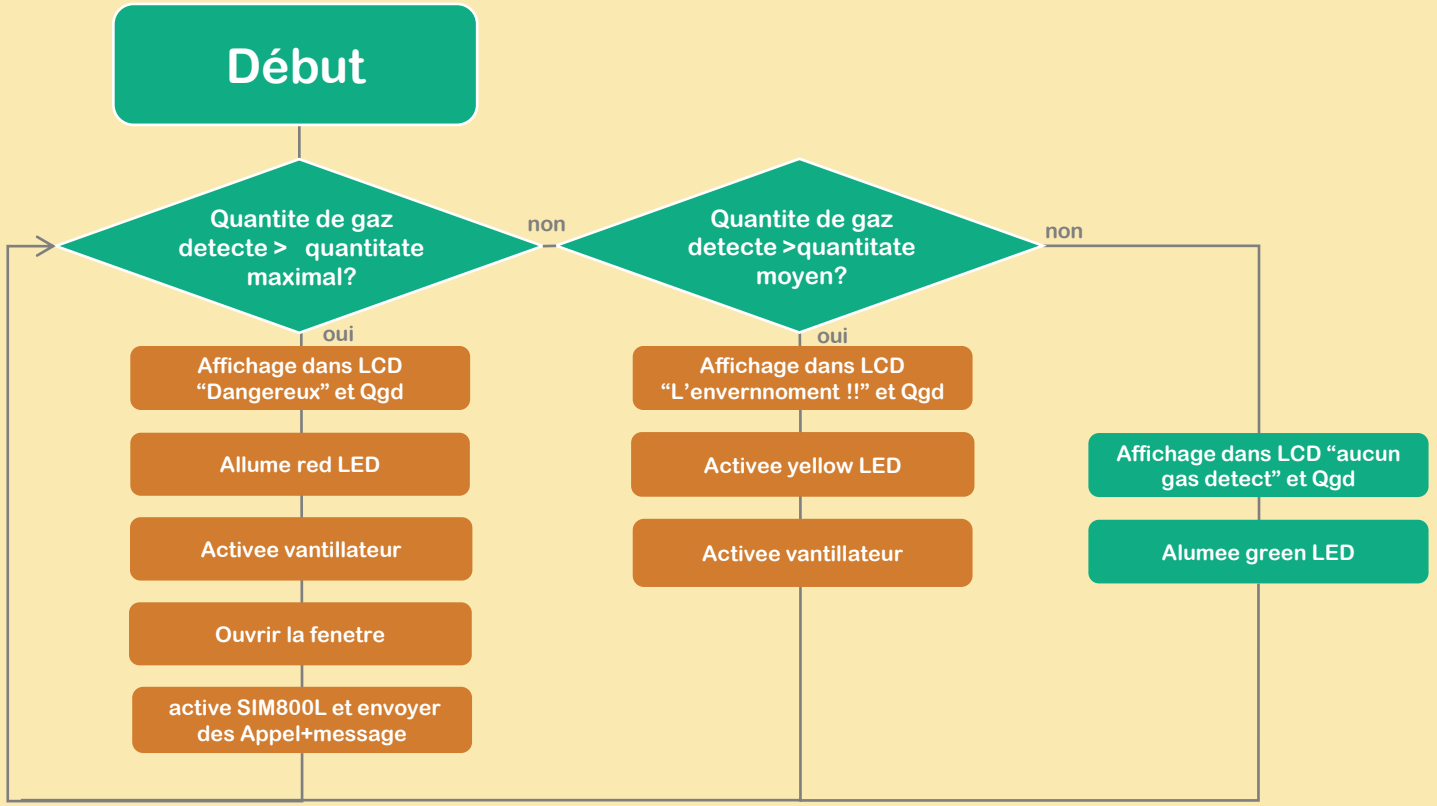
Prototype de l'operation



Algorigramme (fonctionnement)



Prototype de l'operation



Algorigramme (fonctionnement)

Les grand objectif de Smart Detector

1

Détection



2

Informer user

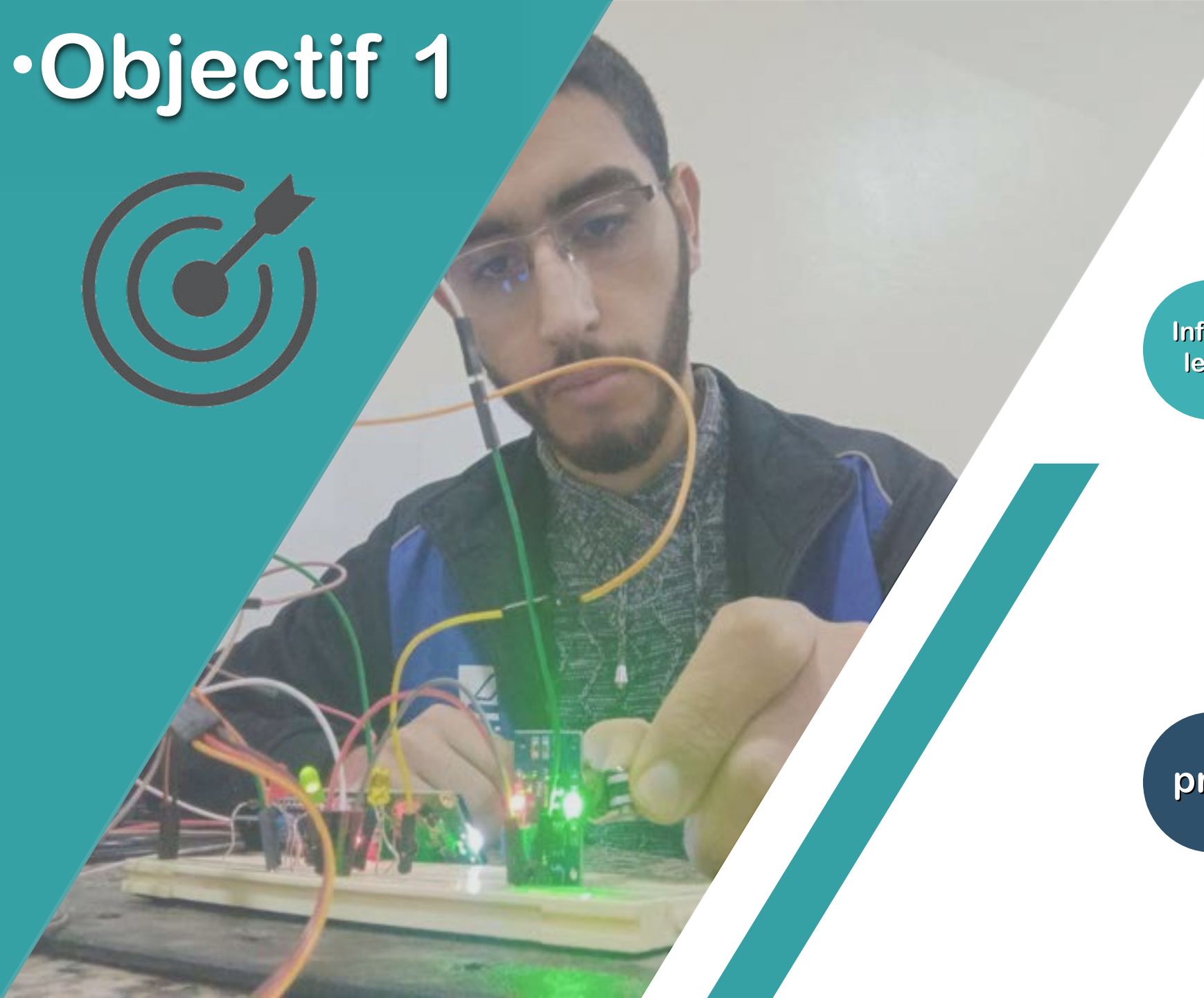


3

Auto Extraction



•Objectif 1



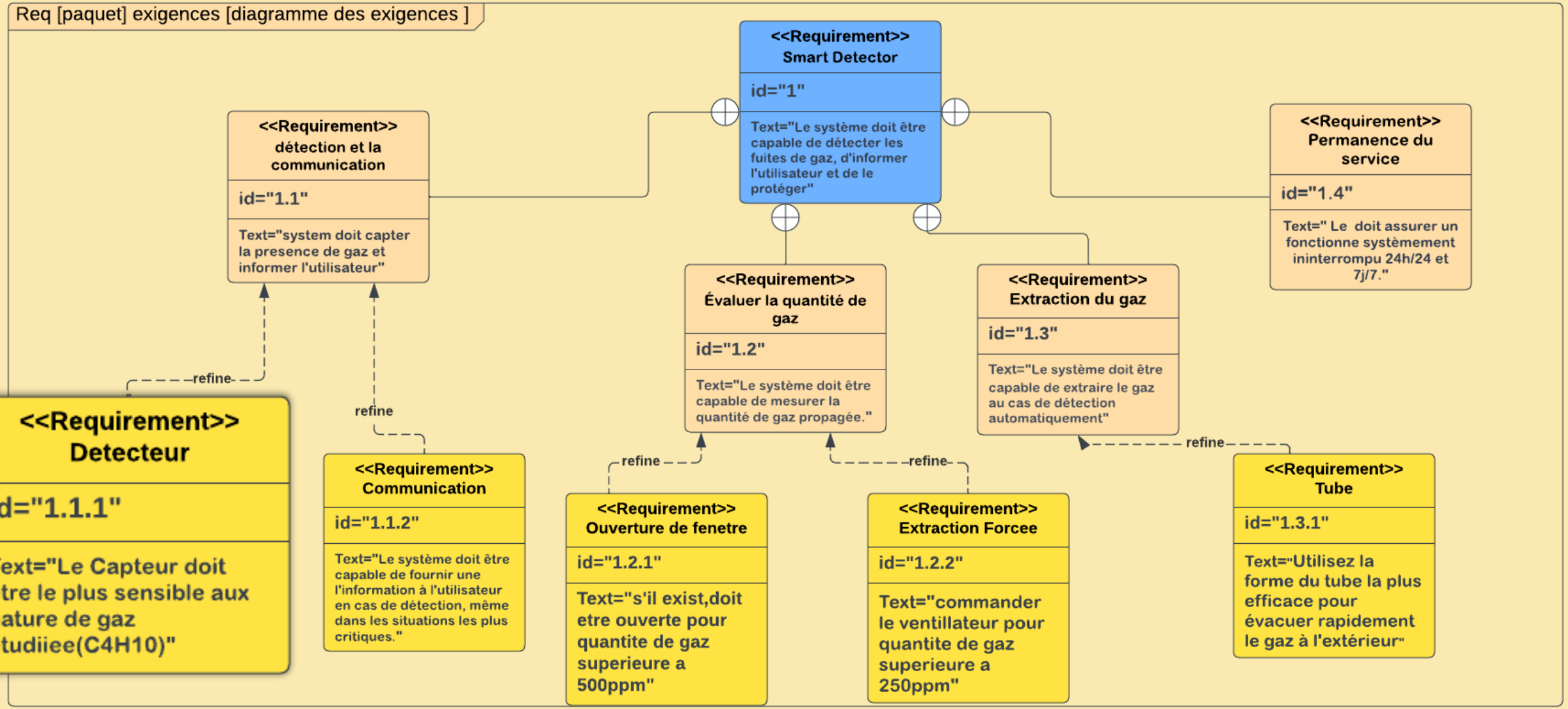
Détection de gaz

01 Information sur le gaz etudiee

02 Choix du capteur

03 prototype

Cahier des charges



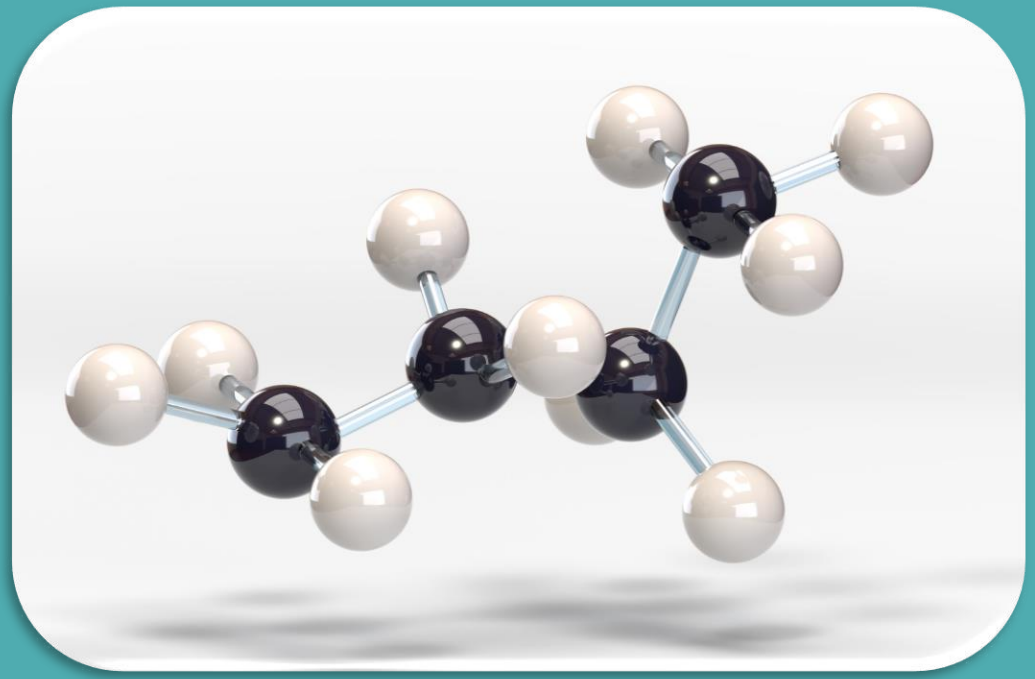


Figure 3 : construction chimique du gaz Butane



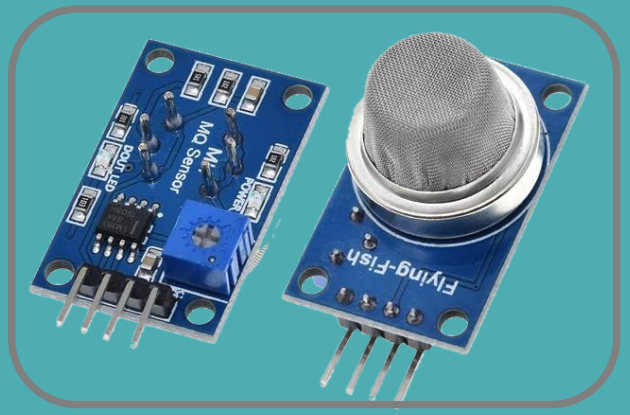
Figure 4 : Bouteille de stockage du gaz

Réaction de combustion : $2C_4H_{10} + 13O_2 \rightarrow 8CO_2 + 10H_2O$

Les capteures

Les performances

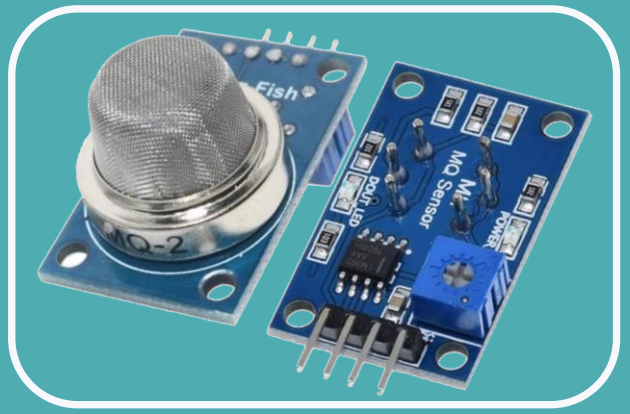
MQ5



Sensibilite aux C_4H_{10}

Sensibilite au (methane,l'alcool)

MQ2



Sensibilite aux C_4H_{10}

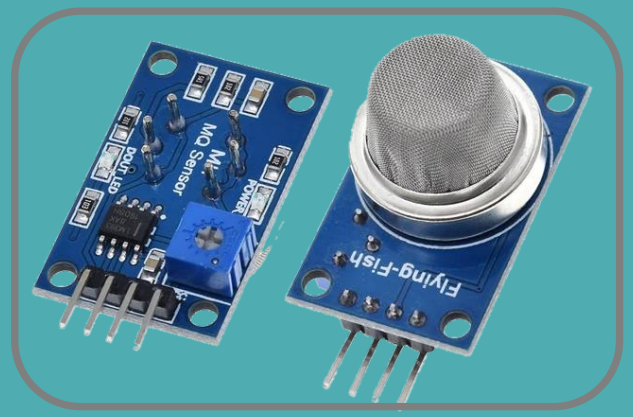
Sensibilite au (methane,l'alcool)

Les capteures

Les performances

syntheses

MQ5

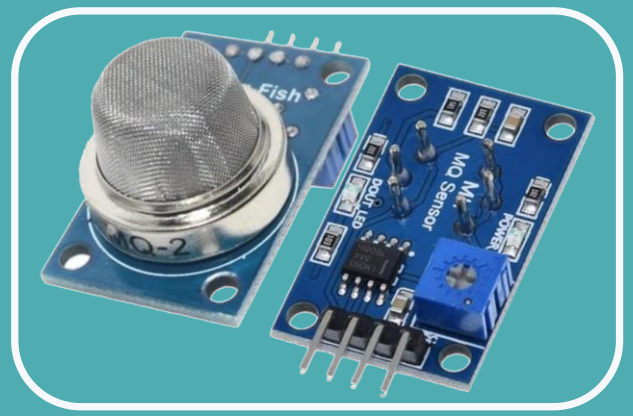


Sensibilite aux C_4H_{10}

Sensibilite au(methane,l'alcool)

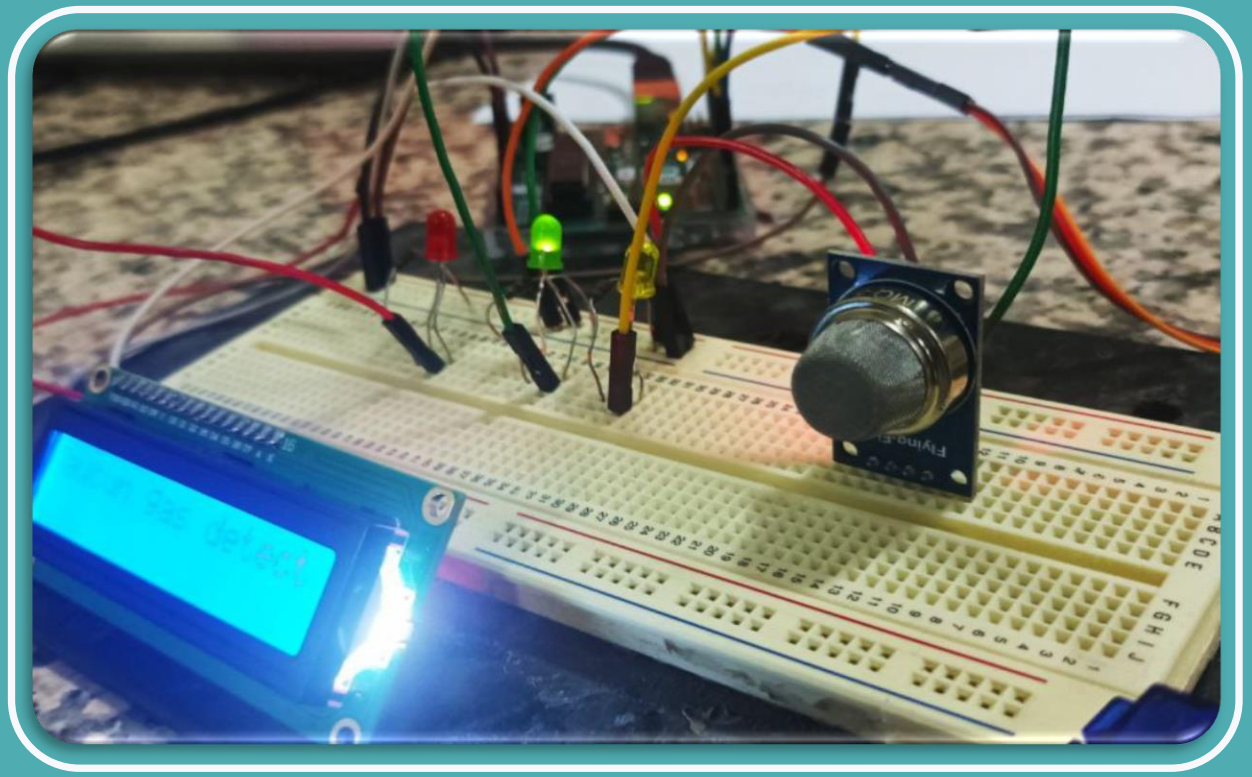
MQ5

MQ2



Sensibilite aux C_4H_{10}

Sensibilite au(methane,l'alcool)



Montage électrique réel

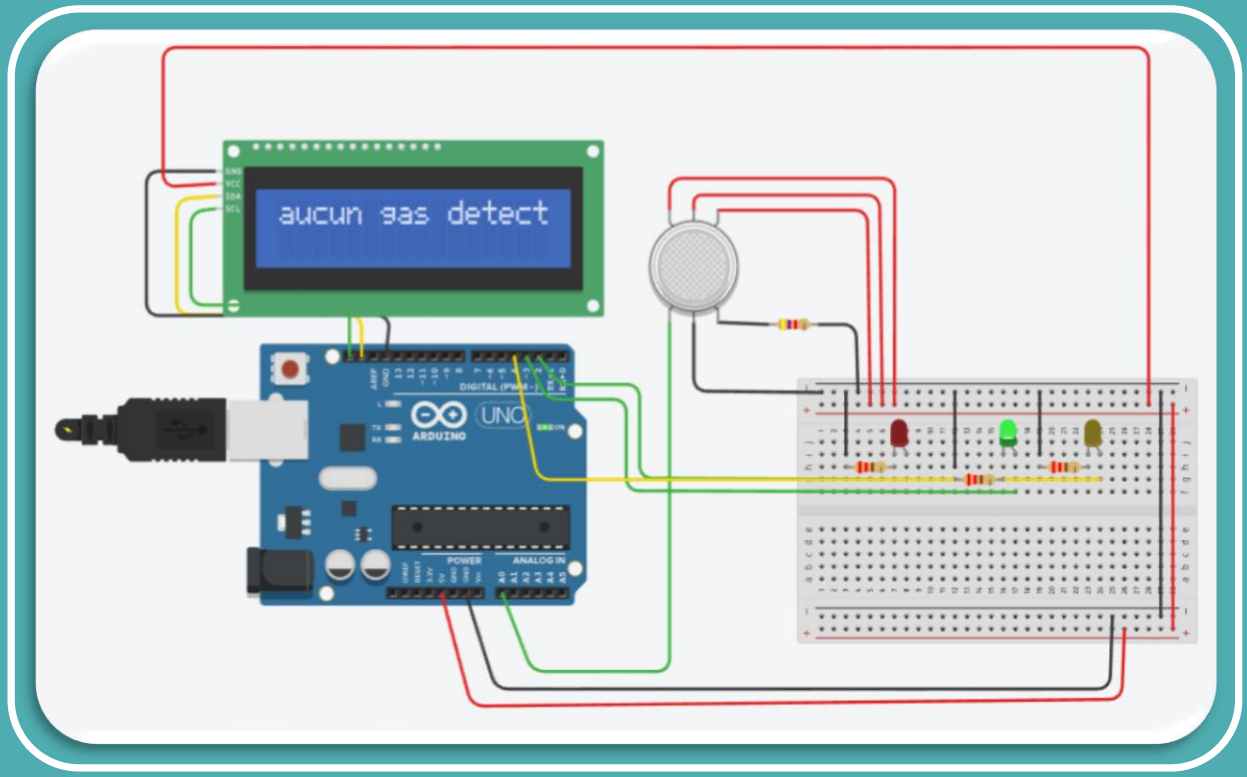
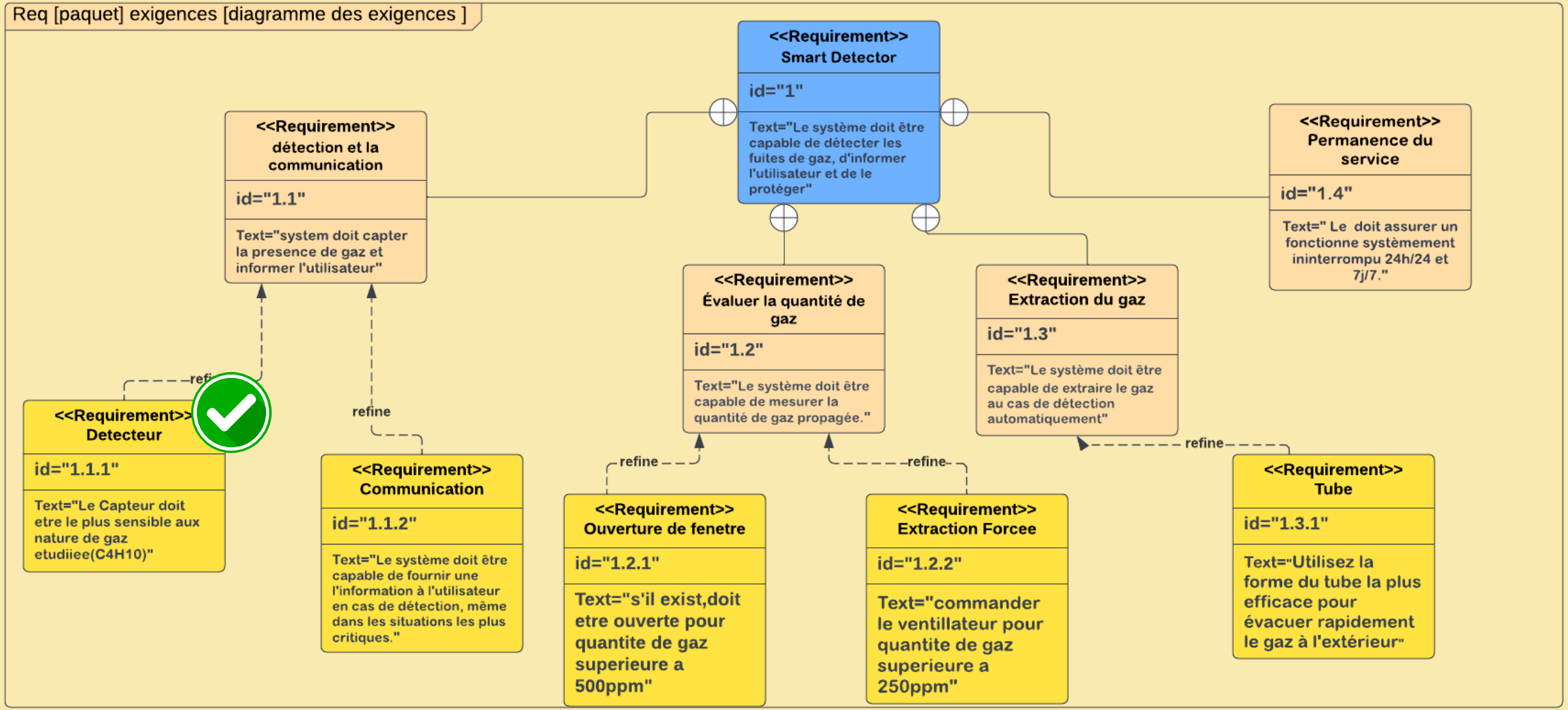


schéma électrique du câblage(Tinkerd)

Cahier des charges



• Objectif 2



Avertissement visuel

01

02

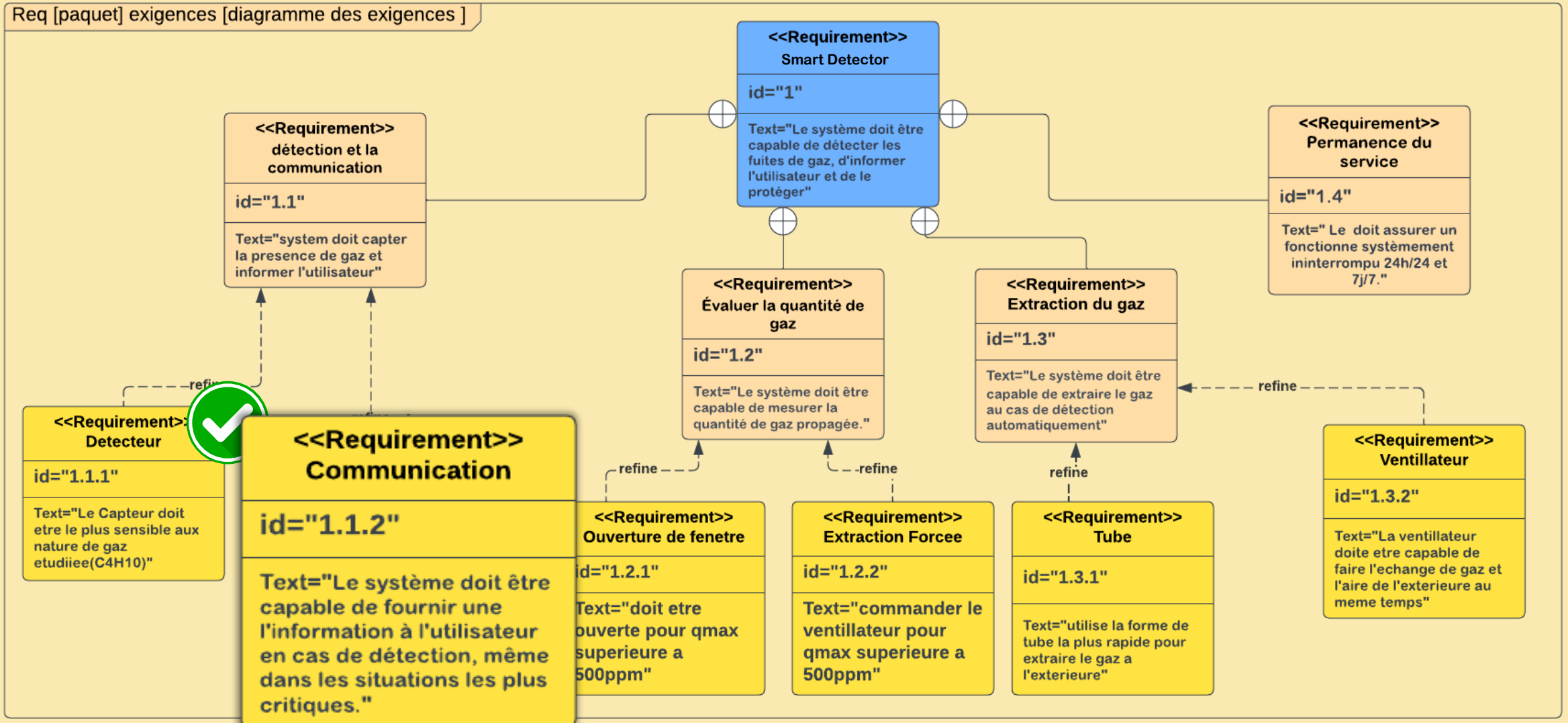
Avertissement par des appeles et messages

prototype

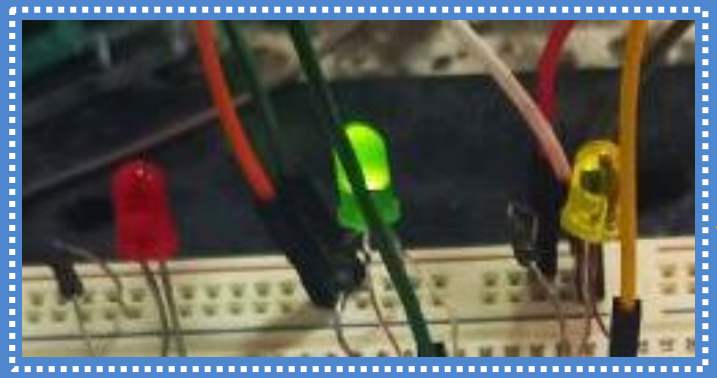
03

Avertissement de l’utilisateur

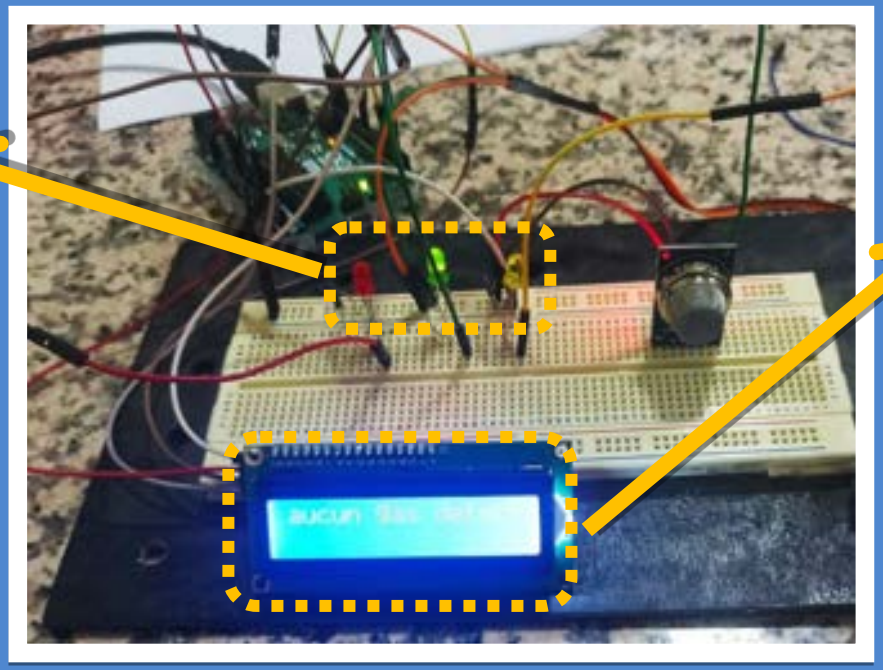
Cahier des charges

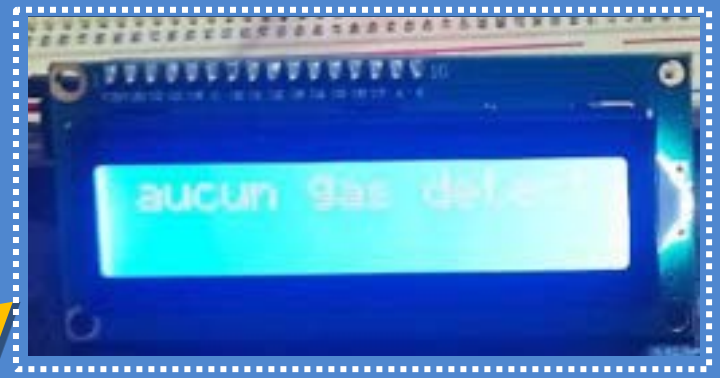


Avertissement visuel



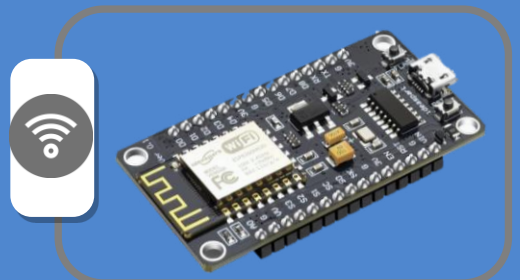
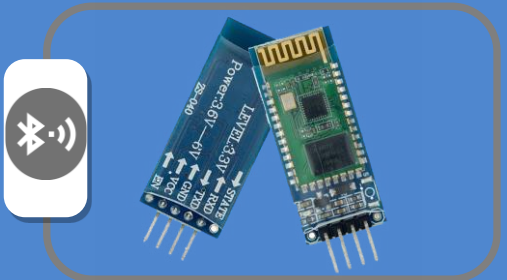
3 LED



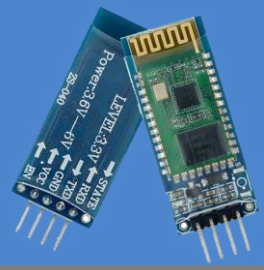


Afficheur LCD

Avertissement par des appelles et messages

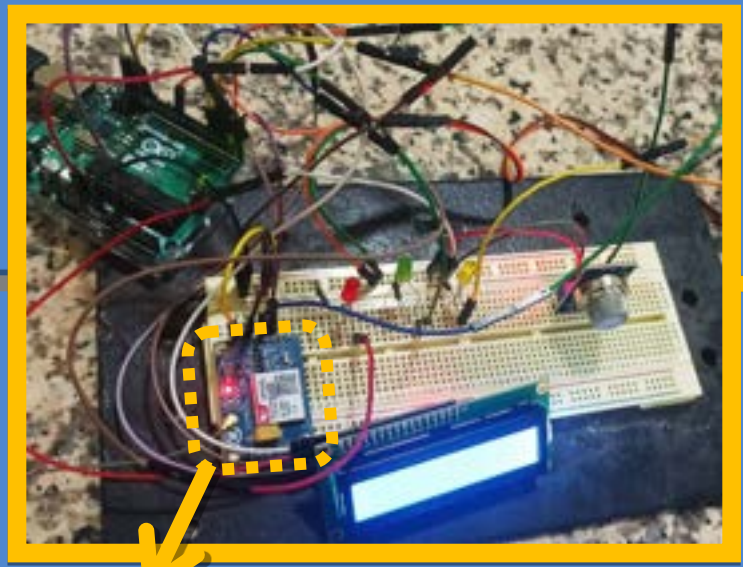
	SIM800L V2,0	ESP8266	HC-05
			
Portée	Pas de limite	Dizaines a qlq (m)	Environ de 10 (m)
Rapidite	Moins rapide	Plus rapide	Rapide
Compatibilite	Avec tout	Avec tout	Selon version

Avertissement par des appelles et messages

	SIM800L V2,0	ESP8266	HC-05
			
Portée	Pas de limite	Dizaines a qlq (m)	Environ de 10 (m)
Rapidite	Moins rapide	Plus rapide	Rapide
Compatibilite	Avec tout	Avec tout	Selon version
Synthese	✓		

Avertissement par des appelles et messages



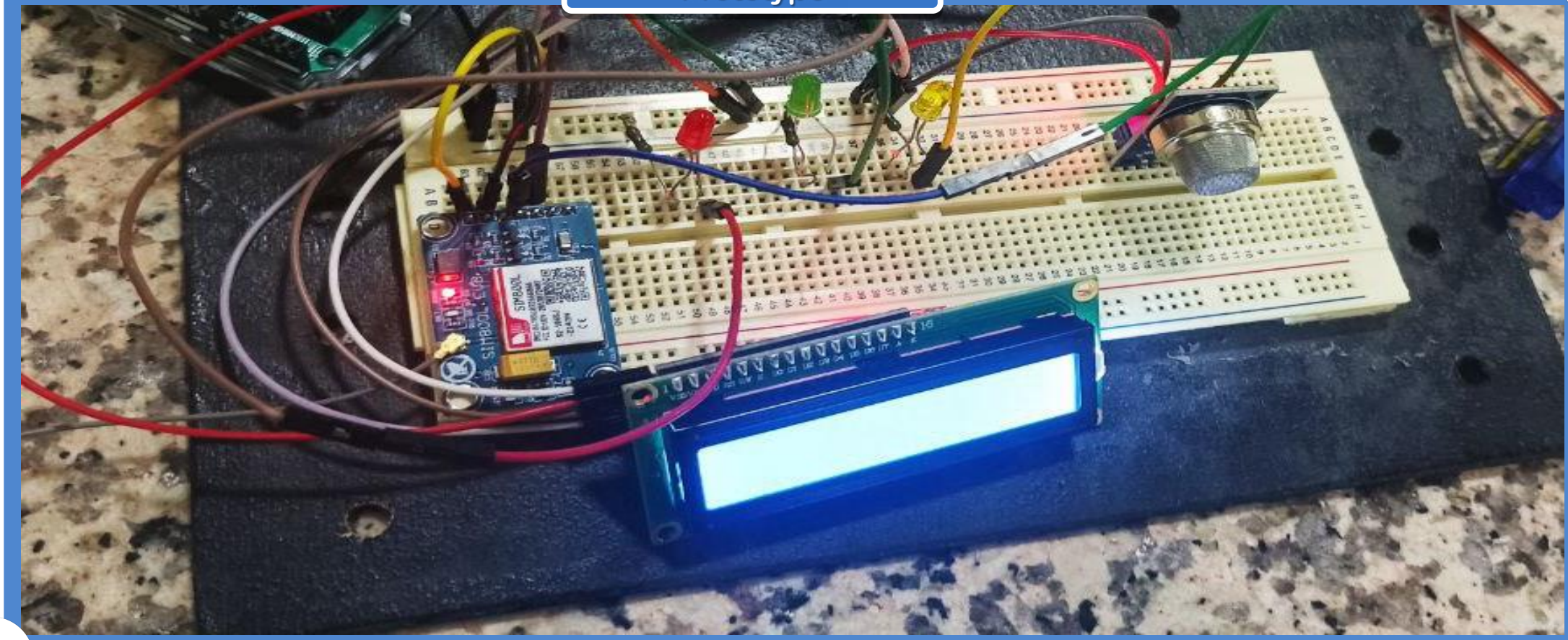




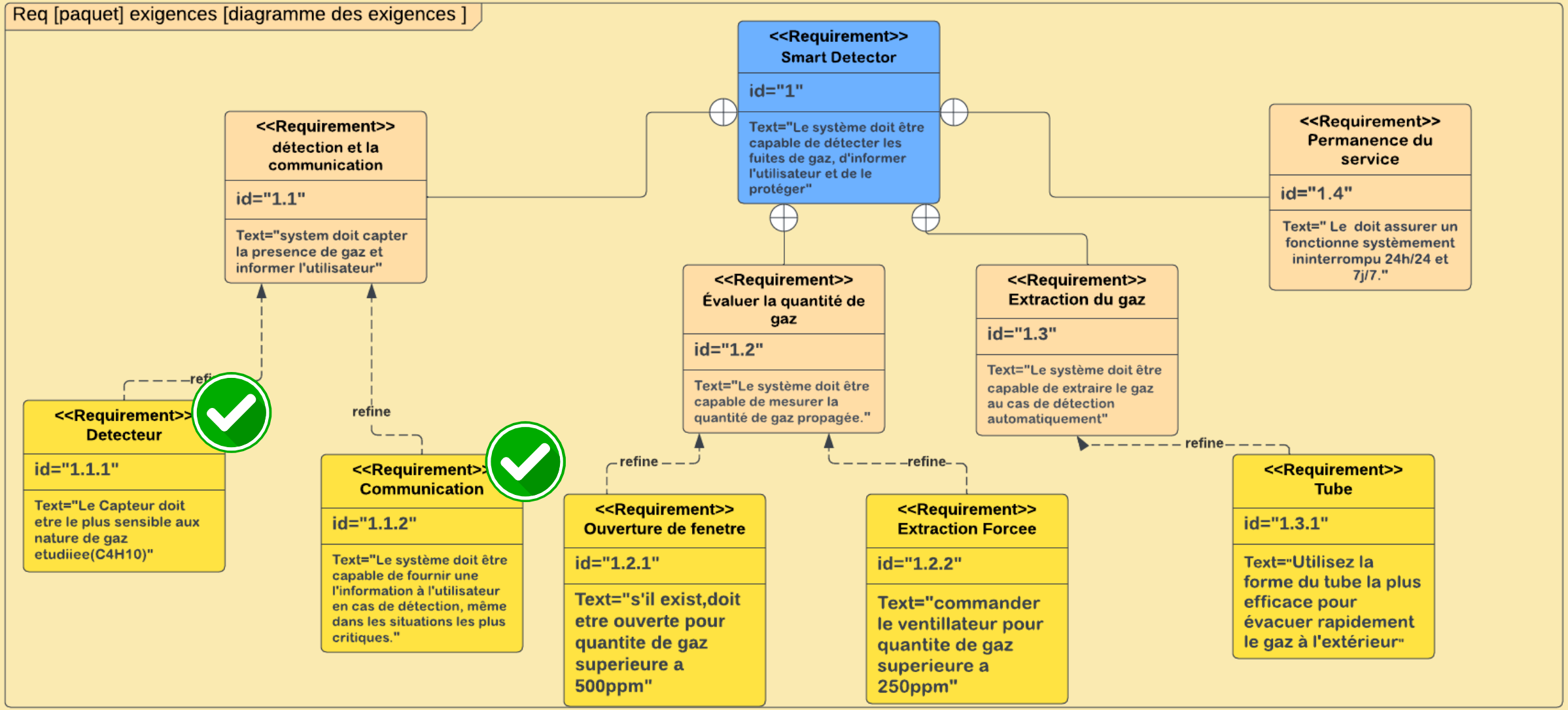


SIM800L V2,0

Prototype



Cahier des charges



Objectif 3

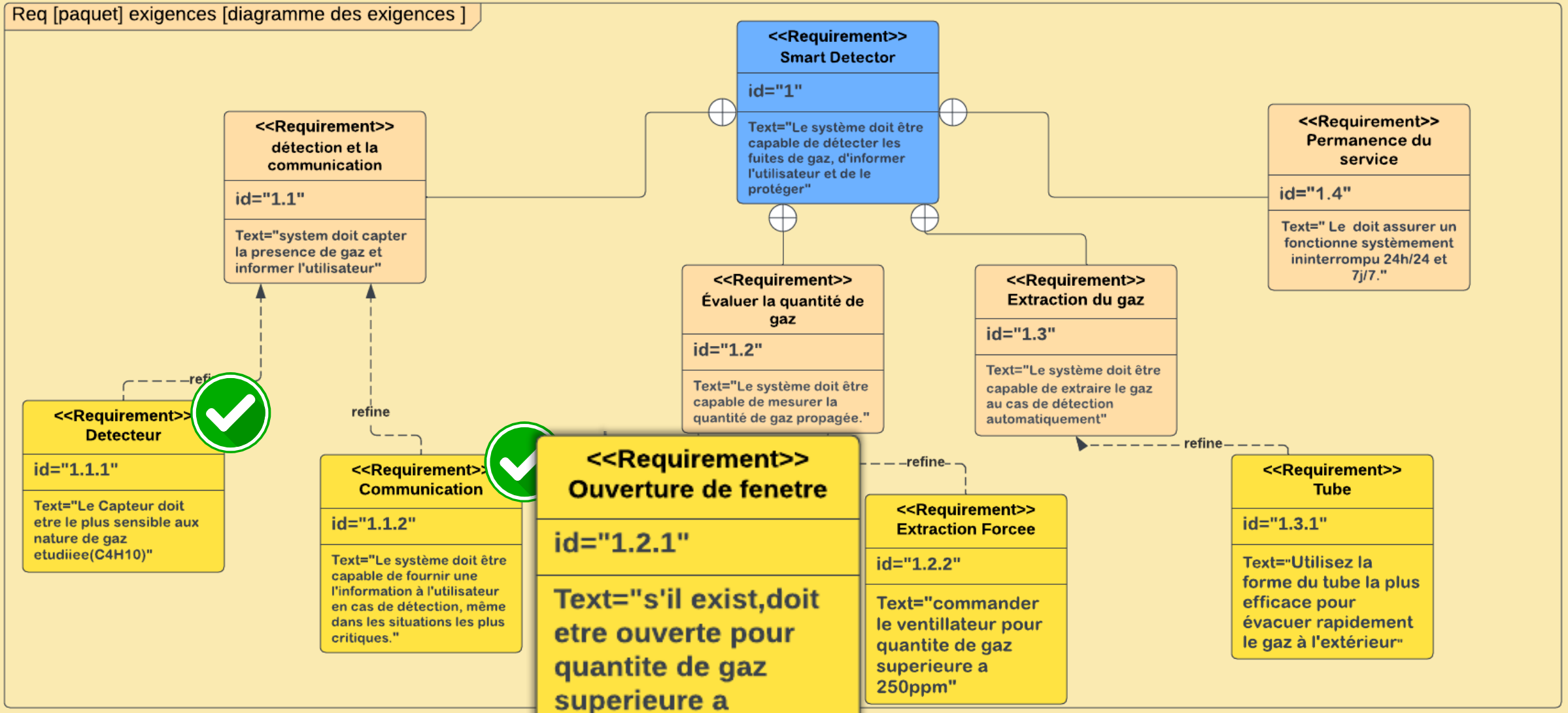


Extraction naturel de gaz par une fenetre



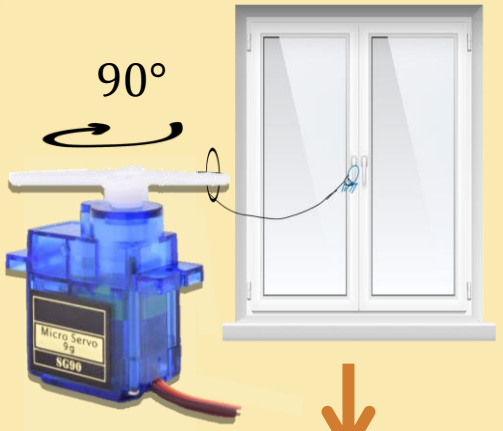


Cahier des charges



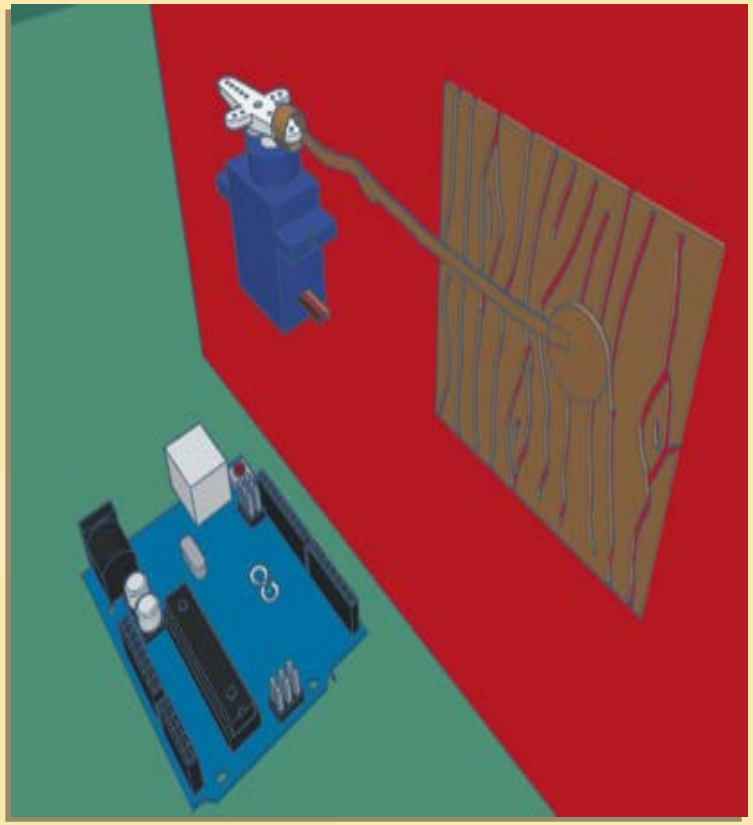


Prototype reel



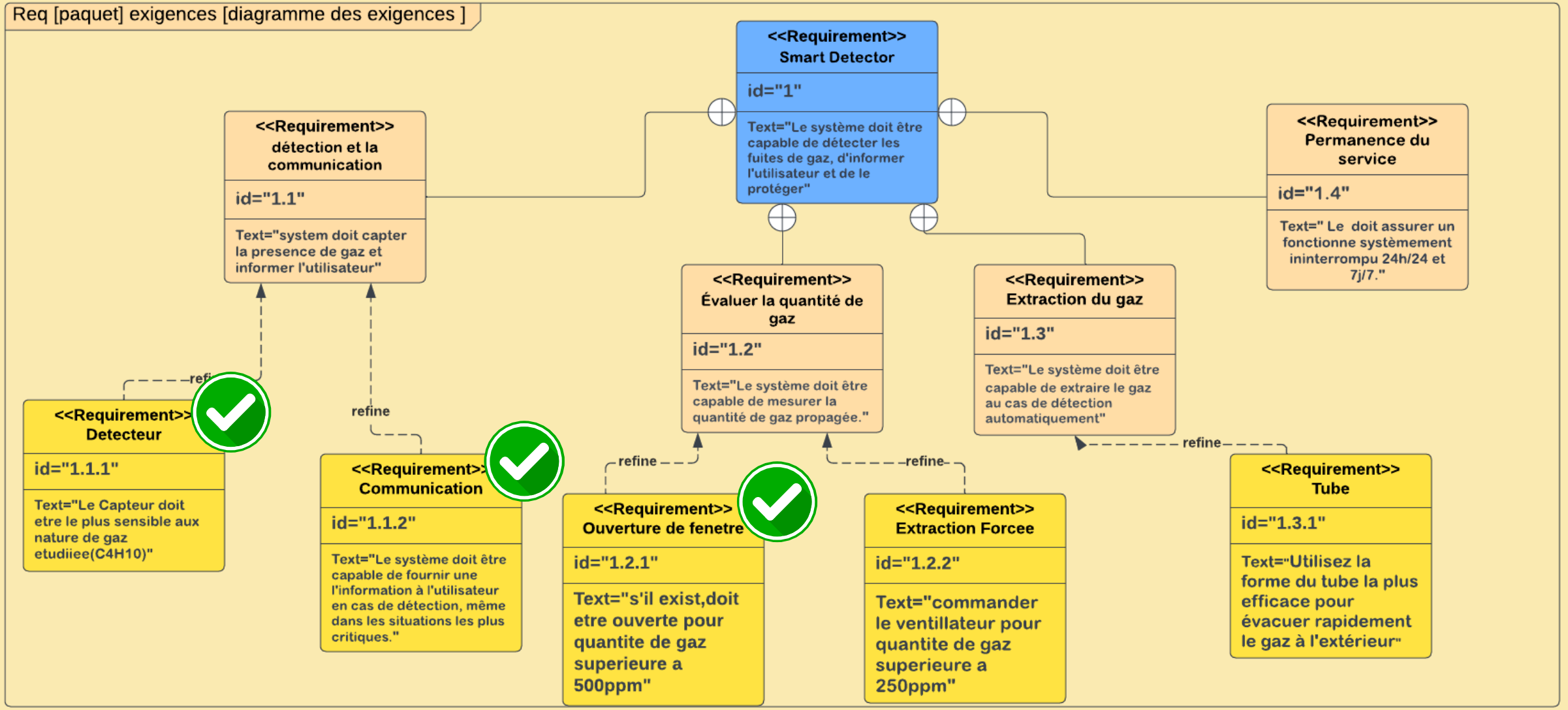
Fermer

Ouvert



Simulation de fonctionnement

Cahier des charges





Est-il suffisant ?

Objectif 4

01 Extration de gaz par une ventillateur

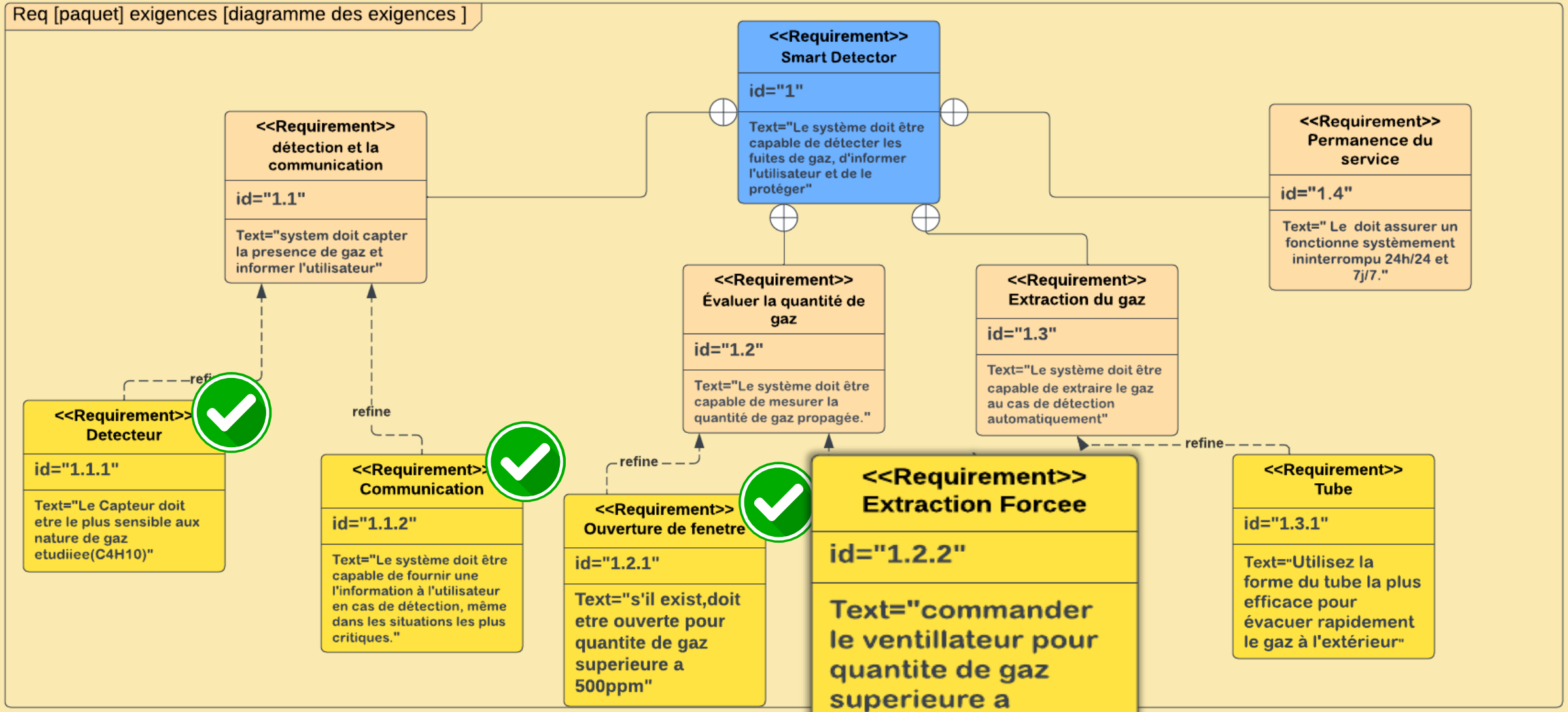
02 Choix forme du tube d'extraction

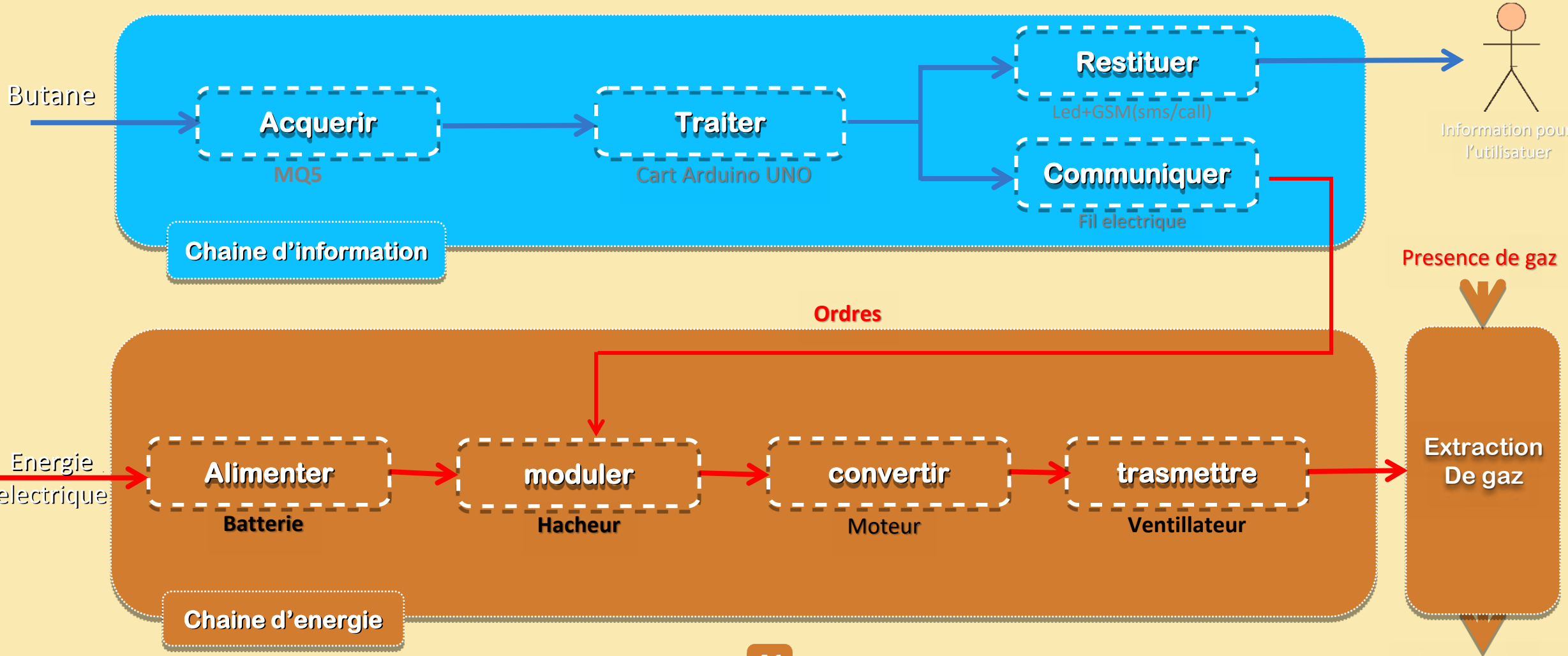
03 Prototype finale



Extraction

Cahier des charges



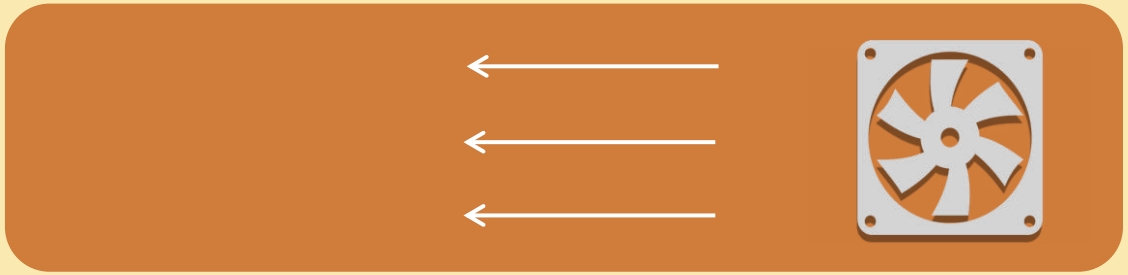

Butane
Acquerir
MQ5
Traiter
Cart Arduino UNO
Restituer
Led+GSM(sms/call)
Communiquer
Fil electrique
Information pour l'utilisateur
Chaine d'information
Presence de gaz
Ordres
Energie electrique
Alimenter
Batterie
moduler
Hacheur
convertir
Moteur
trasmettre
Ventillateur
Extraction De gaz
Absence de gaz
Chaine d'energie



Gaz
A l'interieure

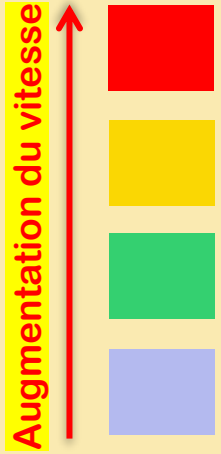
v_0

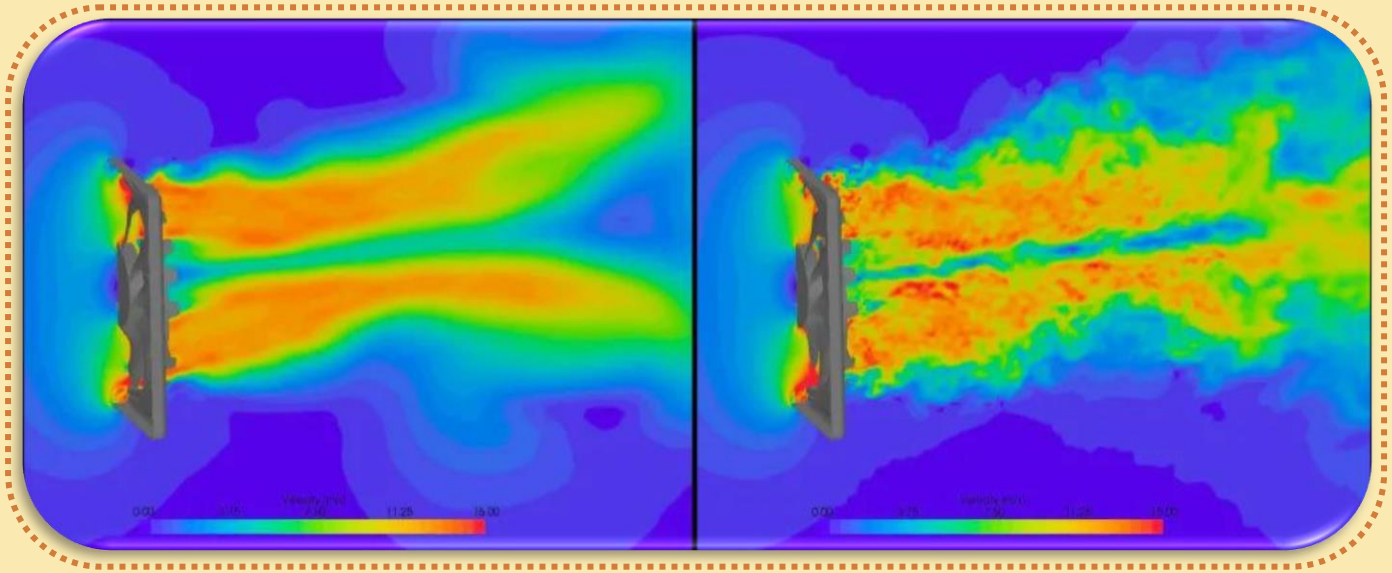
Gaz
A l'exterieure



Gaz
A l'interieure

$v_1 > v_0$

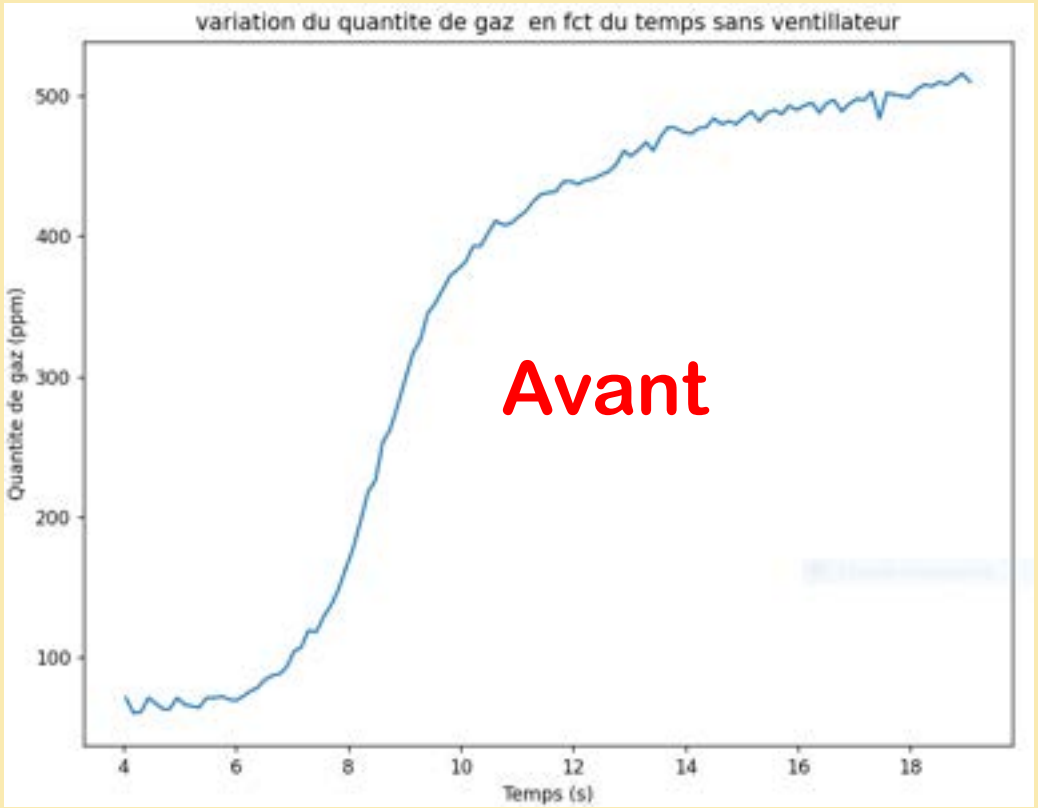




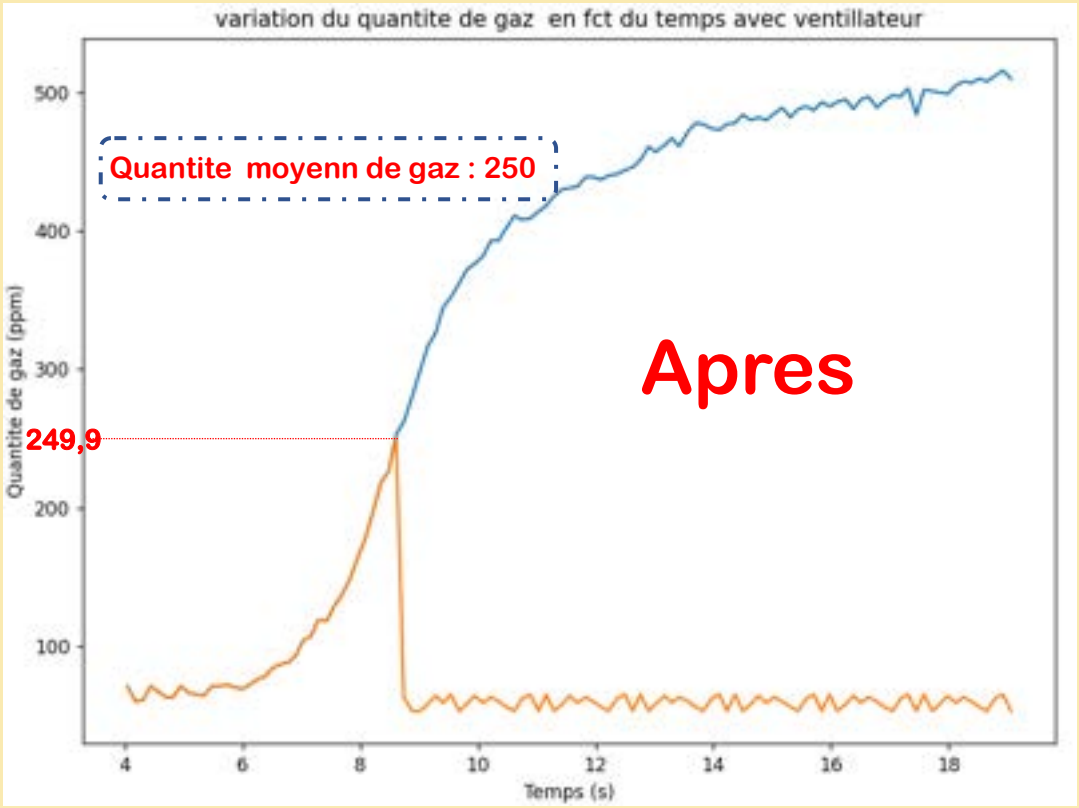
Simulation Vitesse du fluide utilison HELYX

conclusion

D'apres les caracteristique du ventilateur on a une amelioration du Vitesse d'extraction

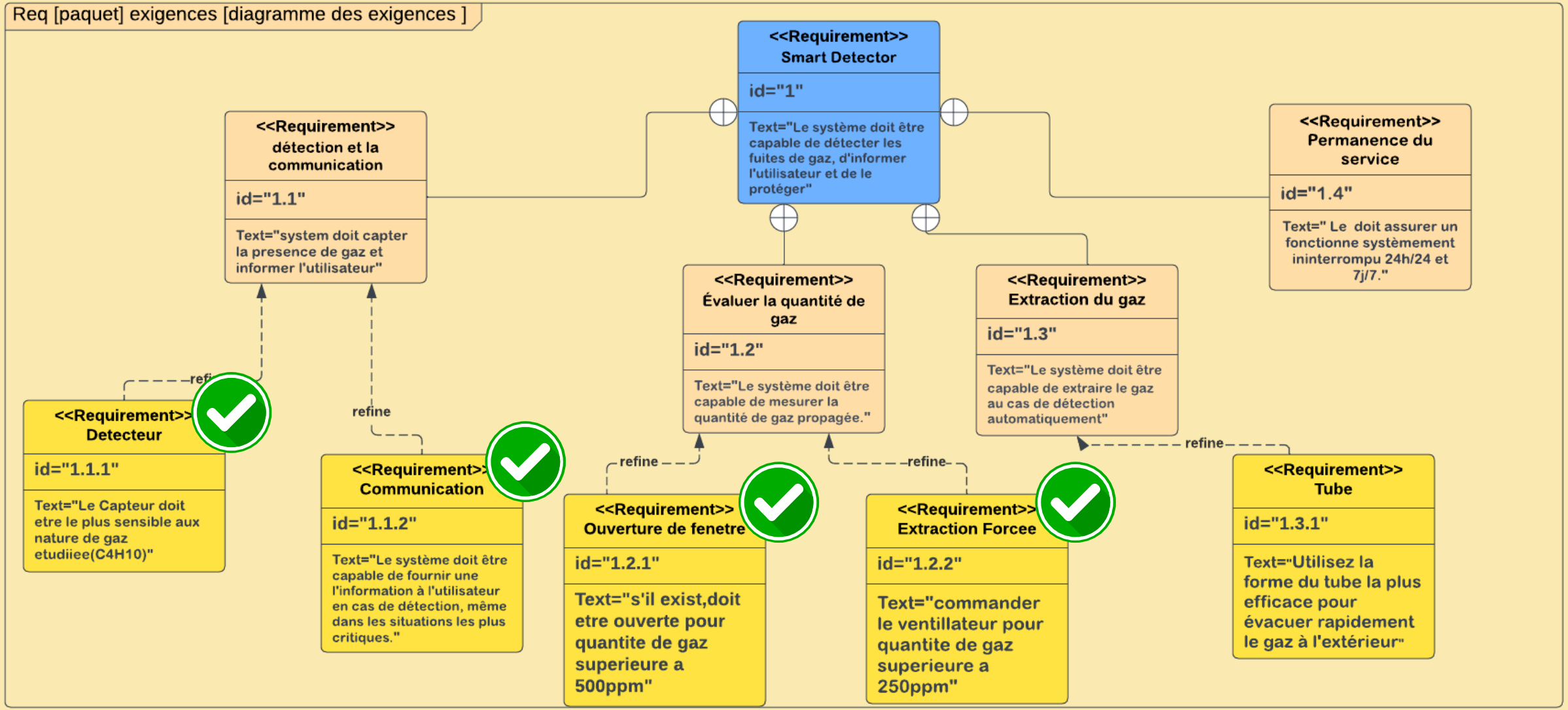






Tracage des Resultats experimental utlison (python)

Cahier des charges





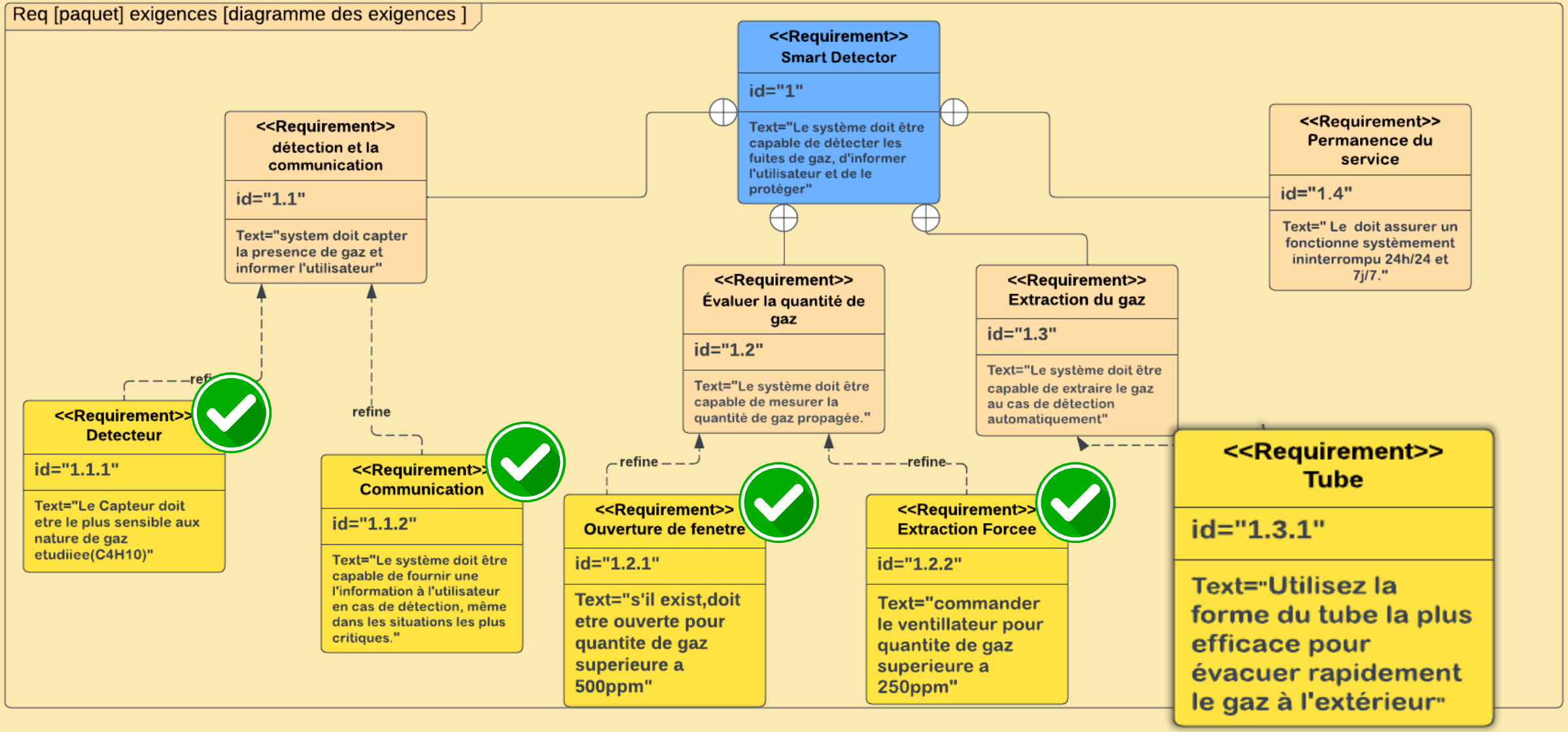
Peut-on l'améliorer plus ?

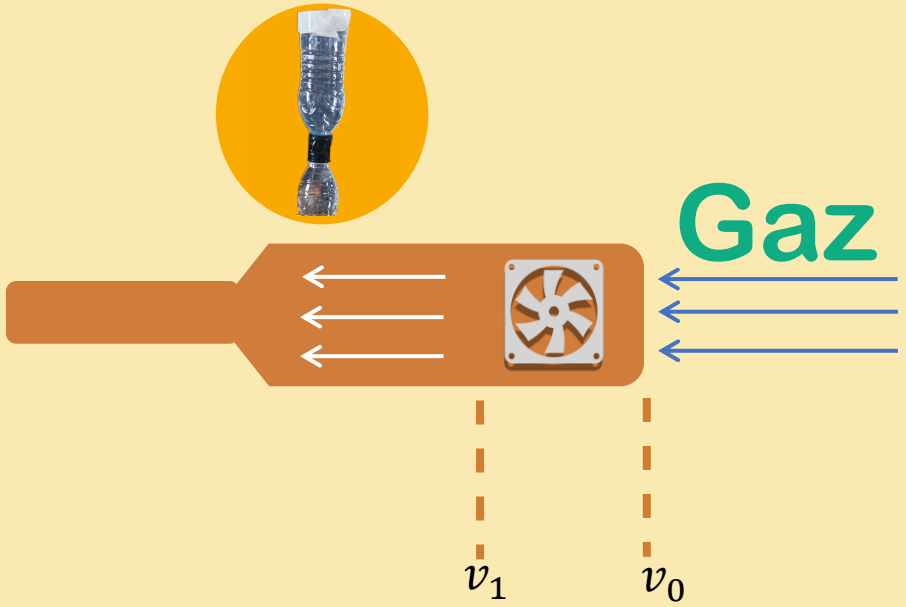


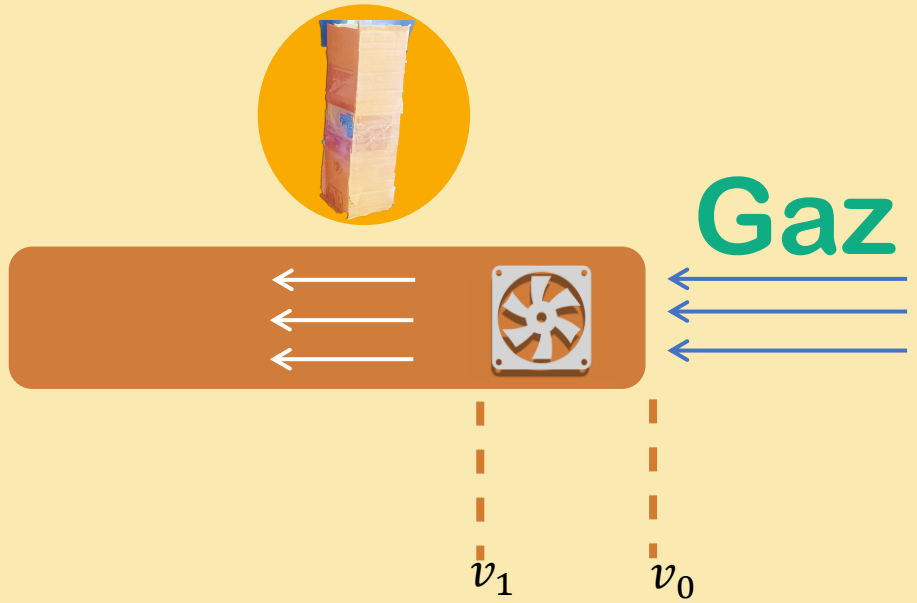


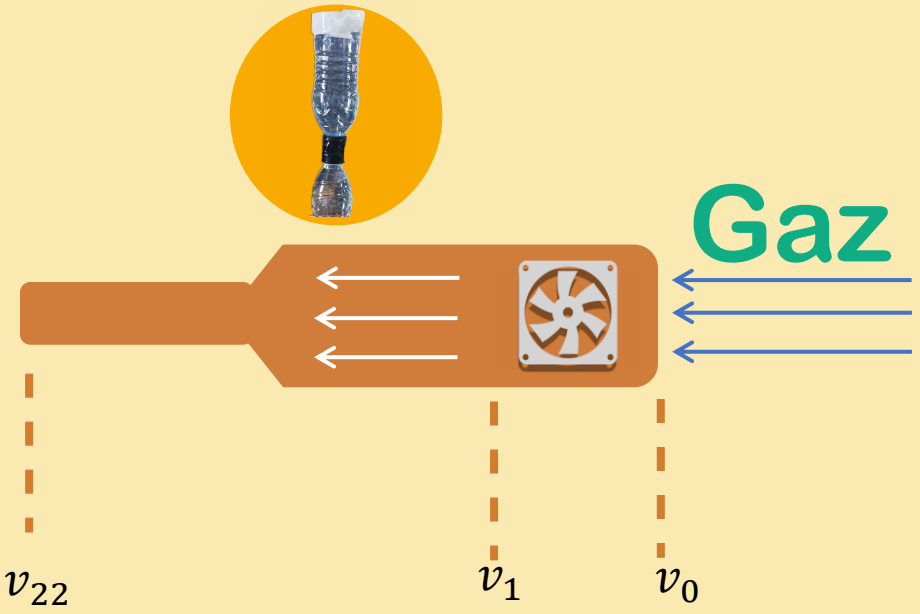


Cahier des charges

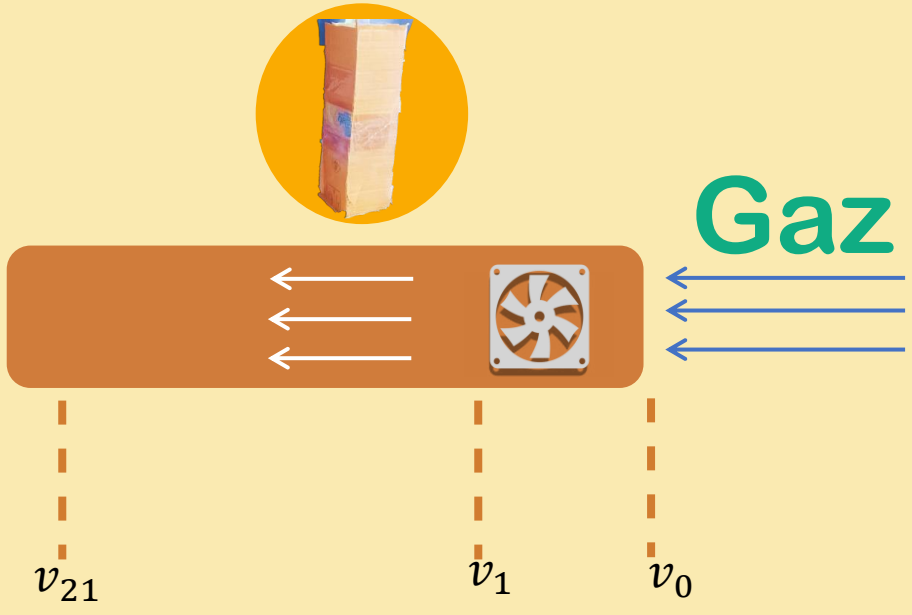



Gaz
v_1
v_0

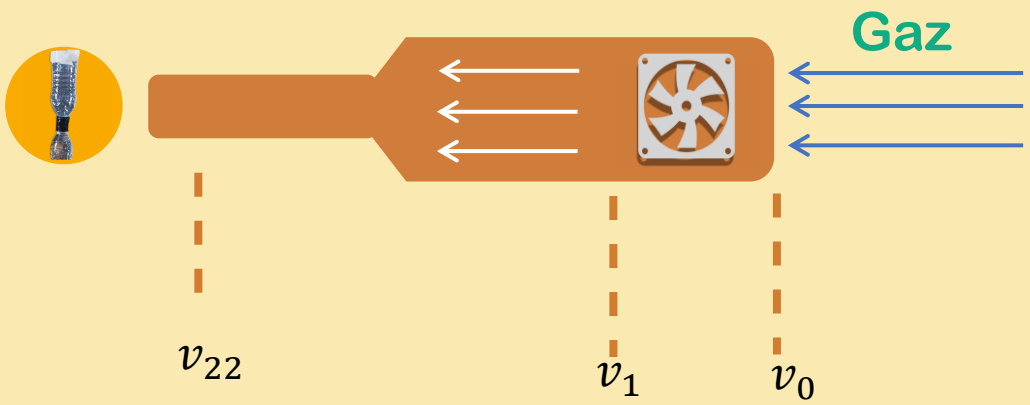

Gaz
v_1
v_0

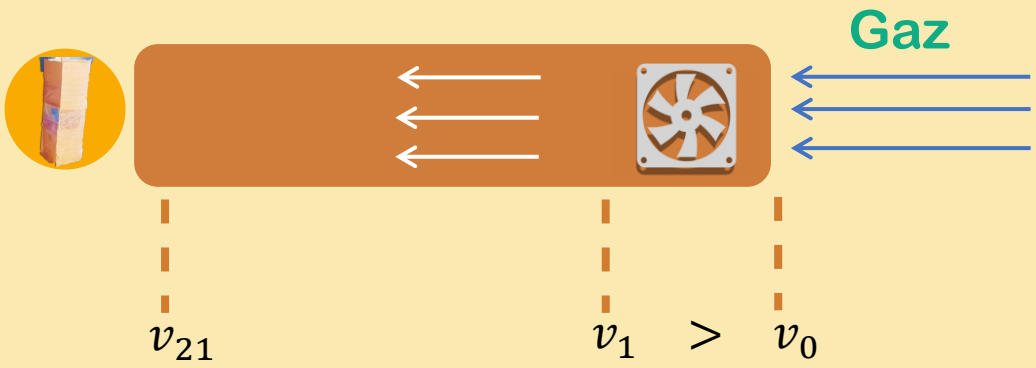






Qu'elle la plus grand v_{22} ou v_{21}





Equation de continuite

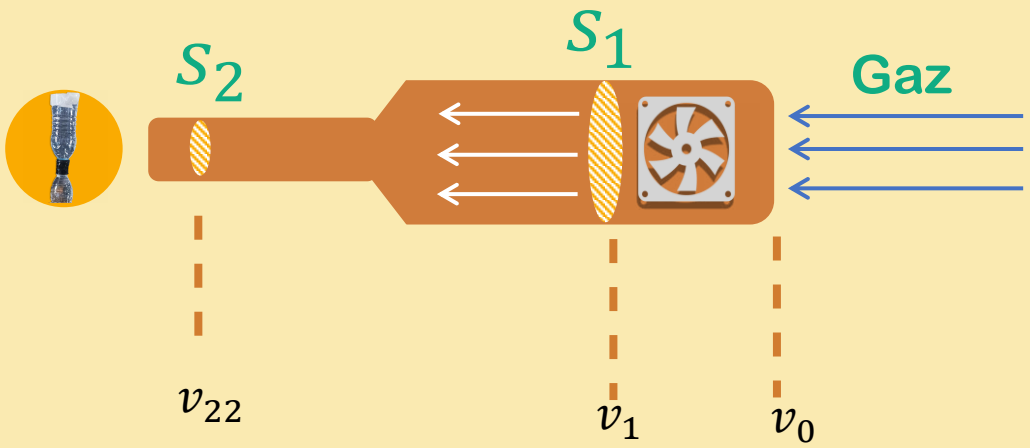
$$s_1 v_1 = s_2 v_2$$

s_1, s_2 : section du tube
v_1, v_2 : vitesses du gaz

Hypothese d'application

Le debit massique est bien constant





Equation de continuite

$$s_1 v_1 = s_2 v_2$$

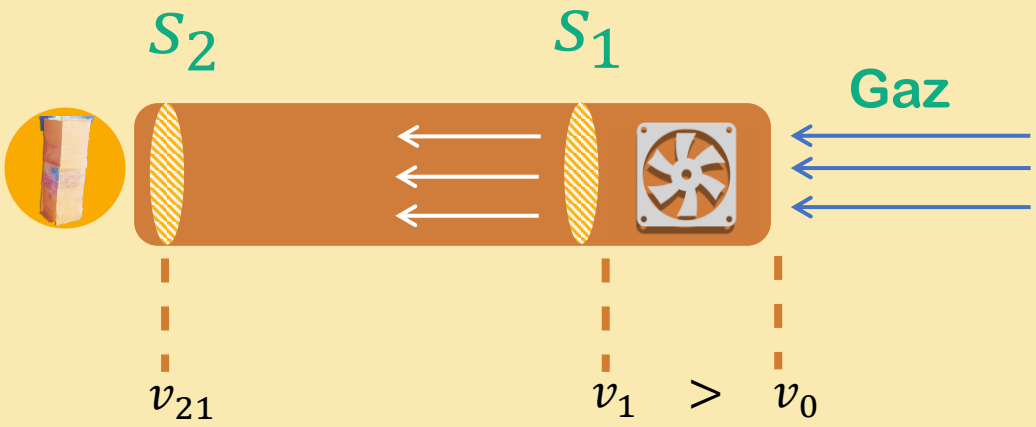
s_1, s_2 : section du tube
v_1, v_2 : vitesses du gaz

Hypothese d'application

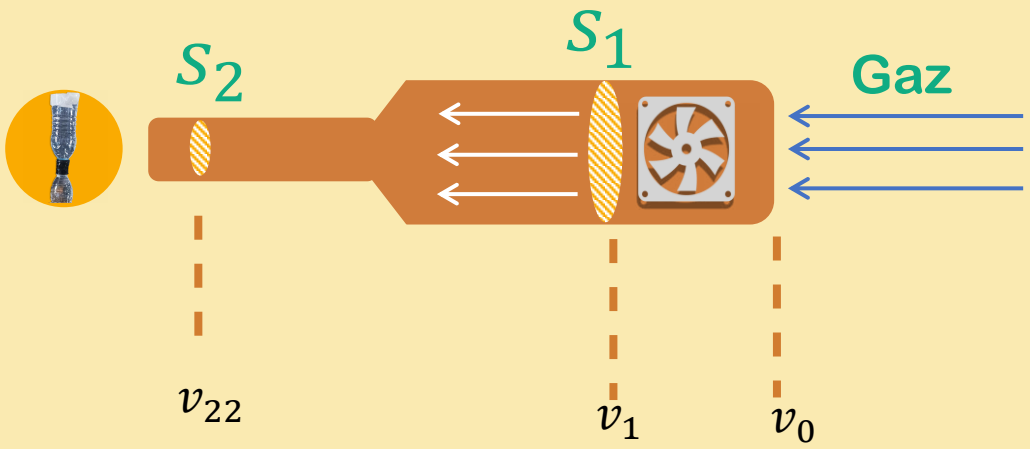
Le debit massique est bien constant



$v_{22} > v_1$



$v_{21} = v_1$



Equation de continuite

$$s_1 v_1 = s_2 v_2$$

s_1, s_2 : section du tube
v_1, v_2 : vitesses du gaz

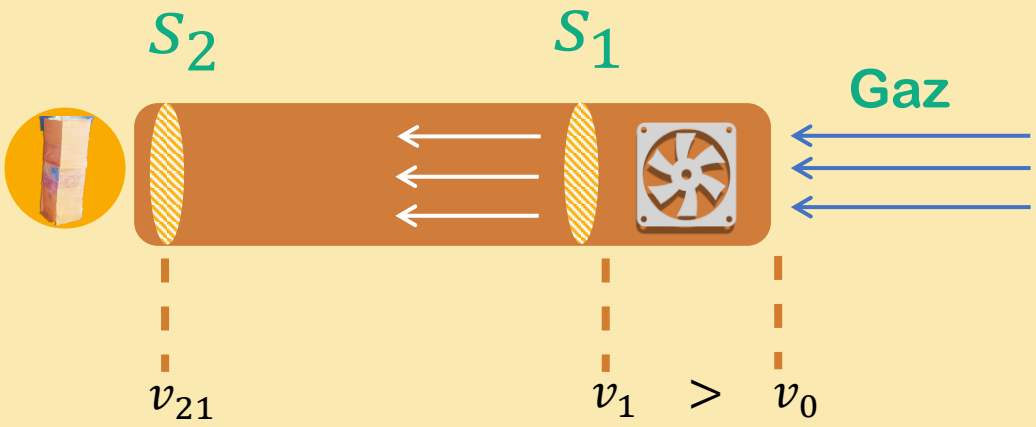
Hypothese d'application

Le debit massique est bien constant



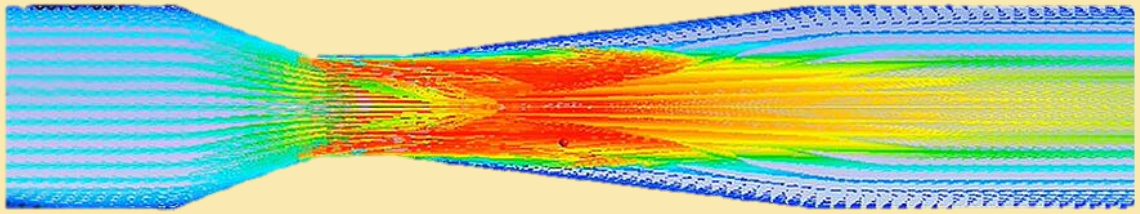
conclusion

$v_{22} > v_1$



$v_{21} = v_1$





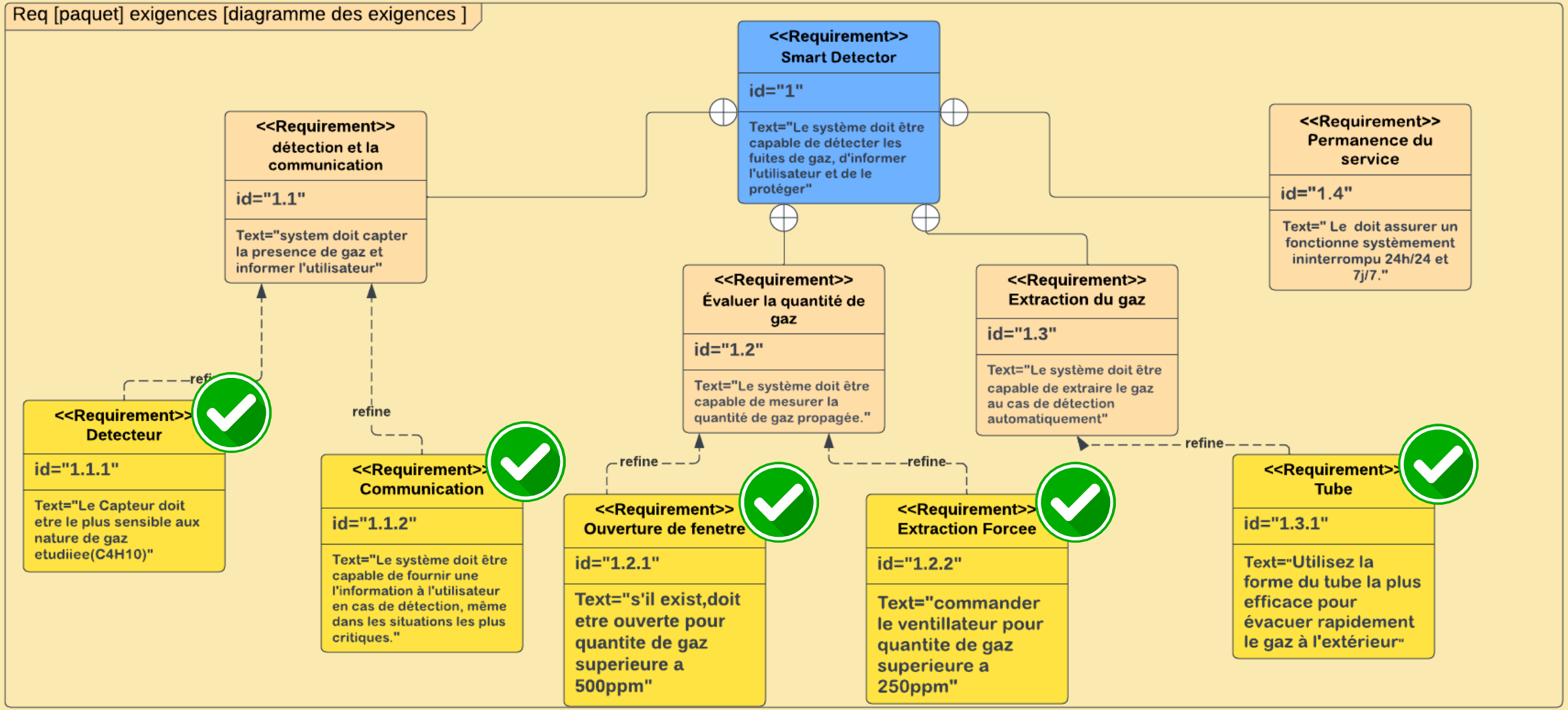




Augmentation du vitesse

Simulation Vitesse du fluide utilison ANSYS

Cahier des charges



Conclusion & Generalisation

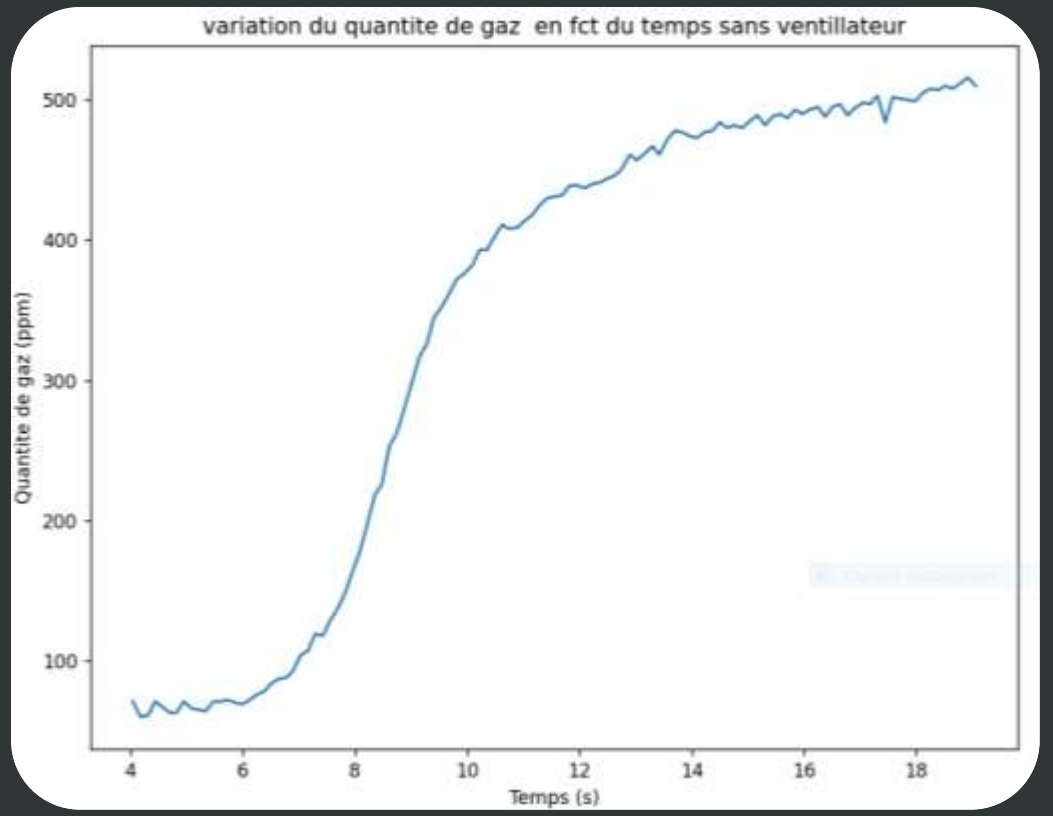


```python
import datetime
import matplotlib.pyplot as plt
valeurs_temps = ['18:29:39.06', '18:29:38.92', '18:29:38.79', '18:29:38.65',
'18:29:38.52',
    '18:29:38.39', '18:29:38.25', '18:29:38.12', '18:29:37.98', '18:29:37.85',
    '18:29:37.72', '18:29:37.58', '18:29:37.45', '18:29:37.31', '18:29:37.18',
    '18:29:37.05', '18:29:36.90', '18:29:36.78', '18:29:36.64', '18:29:36.51',
    '18:29:36.38', '18:29:36.25', '18:29:36.10', '18:29:35.98', '18:29:35.84',
    '18:29:35.71', '18:29:35.58', '18:29:35.44', '18:29:35.31', '18:29:35.17',
    '18:29:35.04', '18:29:34.90', '18:29:34.77', '18:29:34.64', '18:29:34.50',
    '18:29:34.37', '18:29:34.23', '18:29:34.10', '18:29:33.97', '18:29:33.83',
    '18:29:33.70', '18:29:33.57', '18:29:33.42', '18:29:33.30', '18:29:33.17',
    '18:29:33.02', '18:29:32.90', '18:29:32.76', '18:29:32.63', '18:29:32.50',
    '18:29:32.36', '18:29:32.22', '18:29:32.09', '18:29:31.96', '18:29:31.83',
    '18:29:31.69', '18:29:31.56', '18:29:31.43', '18:29:31.29', '18:29:31.16',
    '18:29:31.03', '18:29:30.89', '18:29:30.75', '18:29:30.62', '18:29:30.48',
    '18:29:30.35', '18:29:30.22', '18:29:30.09', '18:29:29.94', '18:29:29.81',
    '18:29:29.68', '18:29:29.54', '18:29:29.41', '18:29:29.28', '18:29:29.15',
    '18:29:29.01', '18:29:28.88', '18:29:28.74', '18:29:28.61', '18:29:28.48',
    '18:29:28.35', '18:29:28.21', '18:29:28.08', '18:29:27.95', '18:29:27.82',
    '18:29:27.69', '18:29:27.56', '18:29:27.43', '18:29:27.29', '18:29:27.16',
    '18:29:27.03', '18:29:26.90', '18:29:26.78', '18:29:26.65', '18:29:26.51',
    '18:29:26.38', '18:29:26.26', '18:29:26.12', '18:29:26.00', '18:29:25.87',
    '18:29:25.74', '18:29:25.61', '18:29:25.48', '18:29:25.34', '18:29:25.22',
    '18:29:25.09', '18:29:24.95', '18:29:24.82', '18:29:24.70', '18:29:24.57',
    '18:29:24.44', '18:29:24.31', '18:29:24.18', '18:29:24.04']
valeurs = [
    510, 516,512, 508, 510, 507, 508, 505, 499, 500, 501, 502, 484, 503, 497, 498,
    494, 489, 497, 495, 488, 495, 493, 490, 493, 487, 490, 488, 482, 489, 485, 480,
    482, 480, 484, 478, 477, 473, 474, 477, 478, 472, 461, 467, 462, 457, 461, 451,
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    402, 393, 393, 382, 376, 372, 362, 352, 345, 326, 317, 298, 280, 262, 253, 226,
    218, 196, 177, 163, 148, 137, 129, 118, 119, 107, 104, 93, 88, 87, 84, 78, 76,
    72, 69, 70, 72, 71, 71, 64, 65, 66, 71, 63, 63, 67, 71, 61, 60, 71]

valeurs_temps_secondes = []
for valeur_temps in valeurs_temps:
    temps = datetime.datetime.strptime(valeur_temps, '%H:%M:%S.%f')
    temps_secondes =temps.second + temps.microsecond / 1e6 -20
    valeurs_temps_secondes.append(temps_secondes)

valeurs_temps_secondes_inverse= valeurs_temps_secondes[::-1]  # Inversion de la
liste temps
valeurs_inverse = valeurs[::-1]  # Inversion de la liste temps

plt.plot(valeurs_temps_secondes_inverse, valeurs_inverse,)
plt.xlabel('Temps (s)')
plt.ylabel('Quantite de gaz (ppm)')
plt.title('variation du quantite de gaz  en fct du temps avec ventillateur')
plt.show()
```



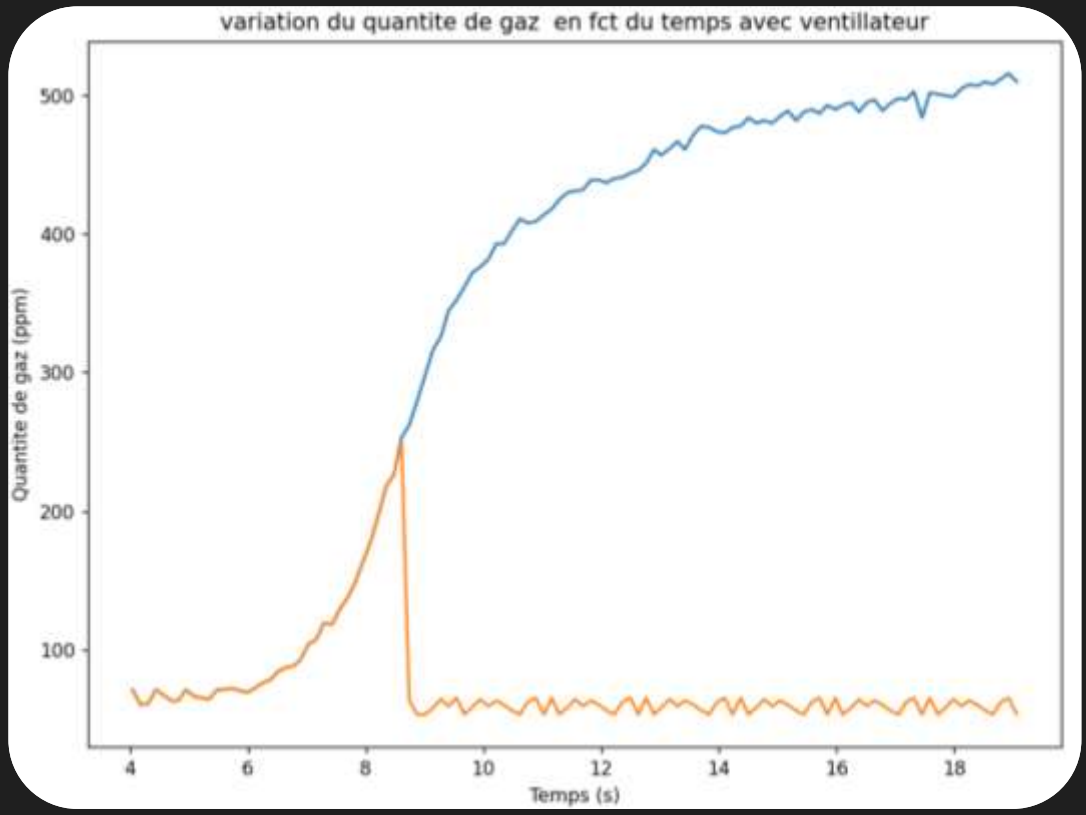
```python
import datetime
import matplotlib.pyplot as plt
valeurs_temps = ['18:29:39.06', '18:29:38.92', '18:29:38.79', '18:29:38.65', '18:29:38.52',
    '18:29:38.39', '18:29:38.25', '18:29:38.12', '18:29:37.98', '18:29:37.85',
    '18:29:37.72', '18:29:37.58', '18:29:37.45', '18:29:37.31', '18:29:37.18',
    '18:29:37.05', '18:29:36.90', '18:29:36.78', '18:29:36.64', '18:29:36.51',
    '18:29:36.38', '18:29:36.25', '18:29:36.10', '18:29:35.98', '18:29:35.84',
    '18:29:35.71', '18:29:35.58', '18:29:35.44', '18:29:35.31', '18:29:35.17',
    '18:29:35.04', '18:29:34.90', '18:29:34.77', '18:29:34.64', '18:29:34.50',
    '18:29:34.37', '18:29:34.23', '18:29:34.10', '18:29:33.97', '18:29:33.83',
    '18:29:33.70', '18:29:33.57', '18:29:33.42', '18:29:33.30', '18:29:33.17',
    '18:29:33.02', '18:29:32.90', '18:29:32.76', '18:29:32.63', '18:29:32.50',
    '18:29:32.36', '18:29:32.22', '18:29:32.09', '18:29:31.96', '18:29:31.83',
    '18:29:31.69', '18:29:31.56', '18:29:31.43', '18:29:31.29', '18:29:31.16',
    '18:29:31.03', '18:29:30.89', '18:29:30.75', '18:29:30.62', '18:29:30.48',
    '18:29:30.35', '18:29:30.22', '18:29:30.09', '18:29:29.94', '18:29:29.81',
    '18:29:29.68', '18:29:29.54', '18:29:29.41', '18:29:29.28', '18:29:29.15',
    '18:29:29.01', '18:29:28.88', '18:29:28.74', '18:29:28.61', '18:29:28.48',
    '18:29:28.35', '18:29:28.21', '18:29:28.08', '18:29:27.95', '18:29:27.82',
    '18:29:27.69', '18:29:27.56', '18:29:27.43', '18:29:27.29', '18:29:27.16',
    '18:29:27.03', '18:29:26.90', '18:29:26.78', '18:29:26.65', '18:29:26.51',
    '18:29:26.38', '18:29:26.26', '18:29:26.12', '18:29:26.00', '18:29:25.87',
    '18:29:25.74', '18:29:25.61', '18:29:25.48', '18:29:25.34', '18:29:25.22',
    '18:29:25.09', '18:29:24.95', '18:29:24.82', '18:29:24.70', '18:29:24.57',
    '18:29:24.44', '18:29:24.31', '18:29:24.18', '18:29:24.04']
valeurs = [
    510, 516,512, 508, 510, 507, 508, 505, 499, 500, 501, 502, 484, 503, 497, 498,
    494, 489, 497, 495, 488, 495, 493, 490, 493, 487, 490, 488, 482, 489, 485, 480,
    482, 480, 484, 478, 477, 473, 474, 477, 478, 472, 461, 467, 462, 457, 461, 451,
    446, 444, 441, 440, 437, 439, 439, 432, 431, 430, 425, 418, 414, 409, 408, 411,
    402, 393, 393, 382, 376, 372, 362, 352, 345, 326, 317, 298, 280, 262, 253, 226,
    218, 196, 177, 163, 148, 137, 129, 118, 119, 107, 104, 93, 88, 87, 84, 78, 76,
    72, 69, 70, 72, 71, 71, 64, 65, 66, 71, 63, 63, 67, 71, 61, 60, 71]

valeur = [ 53, 65, 62, 53, 56, 60, 63, 59, 64, 58, 53, 65,53, 65, 62, 53, 56, 60, 63, 59, 64, 58, 53, 65
    , 53, 65, 62, 53, 56, 60, 63, 59, 64, 58, 53, 65,53, 65, 62, 53, 56, 60, 63, 59, 64, 58, 53, 65,
    53, 65, 62, 53, 56, 60, 63, 59, 64, 58, 53, 65, 53, 65, 62, 53, 56, 60, 63, 59, 64, 58, 53, 65
    ,59, 64, 58, 53, 53, 63, 250, 226,
    218, 196, 177, 163, 148, 137, 129, 118, 119, 107, 104, 93, 88, 87, 84, 78, 76,
    72, 69, 70, 72, 71, 71, 64, 65, 66, 71, 63, 63, 67, 71, 61, 60, 71]

valeurs_temps_secondes = []
for valeur_temps in valeurs_temps:
    temps = datetime.datetime.strptime(valeur_temps, '%H:%M:%S.%f')
    temps_secondes =temps.second + temps.microsecond / 1e6 -20
    valeurs_temps_secondes.append(temps_secondes)

valeurs_temps_secondes_inverse= valeurs_temps_secondes[::-1]  # Inversion de la liste temps
valeurs_inverse = valeurs[::-1]  # Inversion de la liste temps
valeur_inverse = valeur[::-1]

plt.plot(valeurs_temps_secondes_inverse, valeurs_inverse,)
plt.plot(valeurs_temps_secondes_inverse, valeur_inverse)
plt.xlabel('Temps (s)')
plt.ylabel('Quantite de gaz (ppm)')
plt.title('variation du quantite de gaz  en fct du temps avec ventillateur')
plt.show()
```

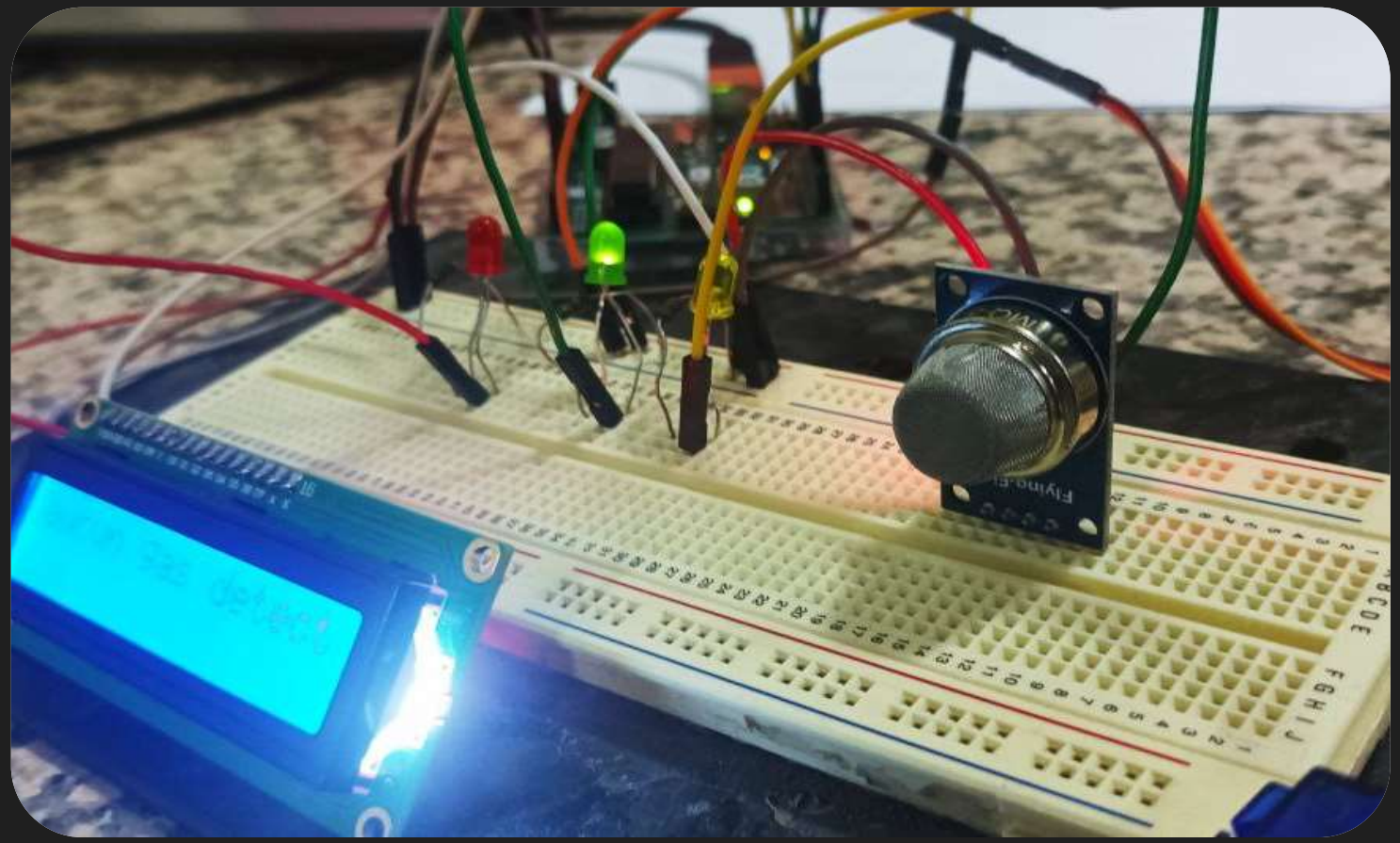


```ino
#include <MQUnifiedsensor.h>
#include <MQ5.h>
#include <LiquidCrystal_I2C.h>

int redled = 2;
int greenled = 3;
int yellowled = 4;
int sensor = A0;
int sensorThresh = 500;
int sensormoyenn = 250;
LiquidCrystal_I2C lcd_1(0x27, 16, 2);

void setup() {
  lcd_1.begin();
  lcd_1.print("    welcome   ");
  lcd_1.setCursor(0, 1);
  lcd_1.print(" to smart detector");
  pinMode(redled, OUTPUT);
  pinMode(greenled, OUTPUT);
  pinMode(yellowled, OUTPUT);
  pinMode(sensor, INPUT);
  Serial.begin(9600);
}

void loop() {
  int analogValue = analogRead(sensor);
  Serial.println(analogValue);
  if (analogValue > sensorThresh) {
    lcd_1.clear();
    lcd_1.print("Dangereux!!!");
    lcd_1.setCursor(0, 1);
    lcd_1.print(analogValue);
    digitalWrite(redled, HIGH);
    digitalWrite(greenled, LOW);
    digitalWrite(yellowled, LOW);
  }
  if (analogValue > sensormoyenn and analogValue < sensorThresh) {
    lcd_1.clear();
    lcd_1.print("l'environnement!!");
    lcd_1.setCursor(0, 1);
    lcd_1.print(analogValue);
    digitalWrite(yellowled, HIGH);
    digitalWrite(greenled, LOW);
    digitalWrite(redled, LOW);
    }
  } else {
    lcd_1.clear();
    lcd_1.print("aucun gas detecte");
    lcd_1.setCursor(0, 1);
    lcd_1.print(analogValue);
    digitalWrite(yellowled, LOW);
    digitalWrite(greenled, HIGH);
    digitalWrite(redled, LOW);
    }
  }
}
```



```ino
#include <MQUnifiedsensor.h>
#include <MQ5.h>
#include <LiquidCrystal_I2C.h>
#include <Servo.h>
#include <SoftwareSerial.h>

SoftwareSerial sim(10, 11);
String number = "+212675105556"; //-> change with your number
int redled = 2;
int greenled = 3;
int yellowled = 4;
int sensor = A0;
int sensorThresh = 500;
int sensormoyenn = 250;
int fan = 11;
int pos = 0;
Servo servo_9;
LiquidCrystal_I2C lcd_1(0x27, 16, 2);

void setup() {
  lcd_1.begin();
  lcd_1.print("    welcome   ");
  lcd_1.setCursor(0, 1);
  lcd_1.print(" to smart detector");
  _buffer.reserve(50);
  sim.begin(9600);
  delay(1);
  servo_9.attach(9, 500, 2500);
  pinMode(redled, OUTPUT);
  pinMode(greenled, OUTPUT);
  pinMode(yellowled, OUTPUT);
  pinMode(fan, OUTPUT);
  pinMode(sensor, INPUT);
  Serial.begin(9600);
}

void loop() {
  int analogValue = analogRead(sensor);
  Serial.println(analogValue);
  if (analogValue > sensorThresh) {
    lcd_1.clear();
    lcd_1.print("Dangereux!!!");
    lcd_1.setCursor(0, 1);
    lcd_1.print(analogValue);
    digitalWrite(redled, HIGH);
    digitalWrite(greenled, LOW);
    digitalWrite(yellowled, LOW);
    digitalWrite(fan, HIGH);
    SendMessage();
    callNumber();

    for (pos = 0; pos <= 90; pos += 1) {
      // tell servo to go to position in variable 'pos'
      servo_9.write(pos);
      // wait 15 ms for servo to reach the position
      delay(1);  // Wait for 15 millisecond(s)
    }
  }
  if (analogValue > sensormoyenn and analogValue < sensorThresh) {
    lcd_1.clear();
    lcd_1.print("l'environnement!!");
    lcd_1.setCursor(0, 1);
    lcd_1.print(analogValue);
    digitalWrite(yellowled, HIGH);
    digitalWrite(greenled, LOW);
    digitalWrite(redled, LOW);
    digitalWrite(fan, HIGH);
    SendMessage();
    callNumber();

    for (pos = 90; pos >= 0; pos -= 1) {
      // tell servo to go to position in variable 'pos'
      servo_9.write(pos);
      // wait 15 ms for servo to reach the position
      delay(1);  // Wait for 1 millisecond(s)
    }
  }
  else {

    lcd_1.clear();
    lcd_1.print("aucun gas detecte");
    lcd_1.setCursor(0, 1);
    lcd_1.print(analogValue);
    digitalWrite(yellowled, LOW);
    digitalWrite(greenled, HIGH);
    digitalWrite(redled, LOW);
    digitalWrite(fan, LOW);
    }
  }

void SendMessage()
{
  //Serial.println ("Sending Message");
  sim.println("AT+CMGF=1");    //Sets the GSM Module in Text Mode
  delay(200);
  //Serial.println ("Set SMS Number");
  sim.println("AT+CMGS=\"" + number + "\"\r"); //Mobile phone number to send message
  delay(200);
  String SMS = "Gas Detect";
  sim.println(SMS);
  delay(100);
  sim.println((char)26);// ASCII code of CTRL+Z
  delay(200);
}

void callNumber() {
  sim.print (F("ATD"));
  sim.print (number);
  sim.print (F(";\r\n"));

}
```



